

Dynamic Policy Development and Implementation: An Electoral Accountability Model

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Abstract

We study a model in which a politician chooses whether and when to implement a policy during a two-period term leading up to an election. Politicians differ in their ability to develop a policy that will produce outcomes voters value. In each period, the politician privately observes the policy’s quality and decides whether to implement it. Once implemented, a well-designed policy may succeed before the election, whereas a badly designed policy never does. Voters observe whether and when the policy is implemented, but not its quality, absent visible success. In equilibrium, competent politicians implement the policy if and only if it is well-designed. Incompetent politicians implement badly designed policy—especially late in their term—to mimic a competent but unlucky politician. A central implication of this behavior is that policies implemented early are, in expectation, of higher quality than those implemented later.

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1 Introduction

Policymaking in representative democracies is inherently dynamic: elected politicians have considerable discretion not only over which policies to implement, but also over when to implement them during their term. Yet the timing of policy implementation reflects more than strategic choice alone. As a presidential staffer quoted in Kingdon (2003) put it:

Just attending to all the technical details of putting together a real proposal takes a lot of time. There's tremendous detail in the work. It's one thing to lay out a statement of principles or a general kind of proposal, but it's quite another thing to staff out all the technical work that is required to actually put a real, detailed proposal together (p. 132).

Thus, developing a policy that will succeed takes both *time* and the *policymaking ability* necessary to resolve technical details, coordinate with bureaucratic actors, build coalitions, and navigate legal or administrative constraints. Ultimately, *whether* and *when* a policy is implemented during a politician's term is a function of a politician's strategic decisions, his policymaking ability, and features of the policymaking process outside of his control.

In this paper, we explore how elections affect the dynamics of policymaking in a setting where there is uncertainty about the ability of an incumbent politician to develop well-designed policies. Key to our analysis are two informational frictions. The first is that voters may not clearly observe the policy development process. While voters may understand a policymaker is interested in implementing a particular policy during his term and can observe whether and when he does, they cannot distinguish whether a delay between the start of the politician's term and policy implementation reflects a politician who is carefully developing a proposal, diligently working out the technical details of the policy, or a politician who lacks a well-designed policy altogether.

For example, following the collapse of its 1994 health care reform effort, the Clinton administration attributed inaction to congressional opposition and institutional constraints

rather than shortcomings in policy design or a failure of leadership (Hacker, 1999). Without directly observing the actual policy development process, voters had a limited ability to evaluate which explanation was correct. A similar problem arises when a politician acts quickly: voters may be unsure whether the politician put sufficient time into developing the policy.

The second informational friction arises after policy implementation: policy success may not be immediately observable. When this is the case, voters may struggle to adjudicate whether an implemented policy that has yet to succeed is well-designed but slow to bear fruit or poorly designed and unlikely to ever succeed. Consequently, assessing the ability of a politician who implemented the policy may be difficult, potentially creating incentives for politicians to request patience when a new policy is enacted but fails to show positive effects.

Such appeals are common. In 1982, for example, Ronald Reagan confronted skepticism about his economic program by acknowledging in a speech to the Oklahoma state legislature that “unemployment remains far too high in too much of the country,” but insisting that:

Returning America to steady economic growth is the answer, not quick fixes.

And that’s what our program is all about, and it will work if we give it time to take hold.¹

In 2010, speaking in Maine one week after signing the Affordable Care Act, Barack Obama similarly cautioned that the public should not expect immediate results. “It will not bring down the cost of health care overnight,” he explained, emphasizing that “over time, costs will come down for families, businesses, and the federal government,” and noting that “it will take about four years to implement this entire plan.”²

Ultimately, what voters infer about a politician based on when he implements a promised policy during his term—and whether he implements it at all—and from policies that suc-

¹Ronald Reagan, “Address Before a Joint Session of the Oklahoma State Legislature, Oklahoma City,” March 16, 1982. The American Presidency Project.

²Barack Obama, “Remarks on Health Care Reform—Portland, Maine,” April 1, 2010. The American Presidency Project.

ceed before the election versus those that do not, depends on how the politician’s strategic behavior depends on these informational frictions. To study this, we develop a dynamic model in which an incumbent politician decides whether and when to implement a policy over the course of a two-period term ending in an election. In each period, the politician privately observes whether a well-designed version of the policy is available. Well-designed policies may generate visible success before the election, while poorly designed policies never do. Politicians differ in their ability to develop well-designed policies: high-ability types sometimes do; low-ability types never do. The politician values reelection and also receives a direct benefit from implementing the policy—say, because the process of implementation helps a political ally—while the voter prefers a well-designed policy to the status quo and the status quo to a badly designed policy.

In our analysis, we focus on the voter’s preferred equilibrium in which high-ability politicians implement the policy immediately upon developing a well-designed version and otherwise delay. Low-ability politicians, by contrast, never develop a well-designed policy and instead randomize across early implementation, late implementation, and inaction. Their objective is to mimic an unlucky high-ability politician—one who either fails to develop a well-designed version of the policy during his term or one who implemented a well-designed policy that failed to produce visible success before the election.

We then examine how low and high-ability politicians’ behavior varies with three features of the policymaking environment: the rate at which high-ability politicians develop viable policies, the likelihood that well-designed policies generate visible success before the election, and the direct benefit politicians receive from implementing the policy. When high-ability politicians are more likely to develop a well-designed policy, they implement it more frequently and earlier in their terms. Because low-ability politicians seek to remain observationally similar to unlucky high types, they also increase their frequency of implementation. By contrast, when policy success becomes more visible, unsuccessful implementation becomes stronger evidence of low ability, especially when the policy is enacted early and therefore has

more time to succeed before the election. Low-ability politicians therefore implement less often and, when they do act, shift implementation later in their term. Increases in the direct benefit of implementation raise the overall frequency of enactment by low-ability politicians, who distribute this additional implementation across early and late periods in fixed proportions in order to remain observationally equivalent to high-ability politicians following unsuccessful outcomes.

These strategic responses generate several substantive implications. First, policies implemented earlier in a politician's term are systematically more likely to be well designed than those implemented later. Second, this quality gap is larger in policy areas where successful outcomes are expected to materialize quickly, since low-ability politicians have stronger incentives to delay implementation when visible success is more likely to appear before the election. Third, stronger direct incentives to implement the policy increase the enactment of badly designed policy and reduce voter welfare, because they make action more attractive even for politicians who are unable to develop effective policy.

The same dynamics also shape what the voter infer about the politician's competence. Policy success, whenever it occurs, fully reveals high ability. By contrast, policies that do not visibly succeed before the election lead voters to draw pessimistic inferences regardless of when they were enacted. Early implementation is initially more favorable because it is more likely to originate from a high-ability politician, but that advantage erodes when visible success fails to materialize. By election day, unsuccessful early and late policies are observationally equivalent, reflecting the low-ability politician's equilibrium strategy of imitating an unlucky high type across policy timing. These inferences also vary systematically with the policy environment: when success is expected to materialize quickly, the absence of success becomes stronger evidence of low ability, whereas stronger private incentives to implement make unsuccessful enactment more damaging and inaction more favorable.

Finally, the analysis considers how these dynamics affect voter and politician welfare. Voter welfare declines when politicians face stronger direct incentives to implement the policy

and when successful outcomes are less likely to become visible before the election, because both forces increase the enactment of badly designed policy. Politicians' welfare moves differently. High-ability politicians benefit from environments rich in opportunities to develop a well-designed policy and in which successful outcomes are likely to become visible, while low-ability politicians prefer environments in which success is rare and policy windows are scarce, making it easier to mimic an unlucky high type.

2 Related Literature

This paper contributes to the formal literature on electoral accountability, which studies how elections discipline politicians under informational frictions (e.g., Canes-Wrone, Herron and Shotts, 2001; Maskin and Tirole, 2004; Ashworth and Shotts, 2010). This literature provides a rich account of how electoral incentives shape *which* policies politicians choose, but largely abstracts from how electoral incentives shape *when* a politician chooses to act. We focus on that question, showing how a politician's desire to win reelection affects whether and when he acts in a setting where there is uncertainty about how long it takes him to develop a well-designed policy and for a well-designed policy to yield a visible success once implemented.

While there is relatively little work on how elections affect when a politician chooses to act, we are not alone in studying this question. In Gibbs (2025), which studies a model that unfolds over two periods, there are two available policy options, and there is uncertainty about when an incumbent politician receives an informative private signal about a policy-relevant state of the world, the precision of which depends on his ability. We emphasize the time required to generate a well-designed policy rather than uncertainty about which option from a menu is correct. In Gieczewski and Li (2022), a politician chooses when to implement a policy of uncertain quality. In their paper and ours, politicians' timing choices directly shape voter information about the quality of the policy because policy effects take time to be observed, but their model does not feature the asymmetric information about incumbent

competence or policy quality that drives our results. Bernheim and Bodoh-Creed (2020) also study policy timing as a deliberate political choice, and in both their paper and ours, earlier action is electorally advantageous, at least initially. In their model, that advantage comes from voters directly rewarding decisiveness. In our model, it comes from the fact that high-ability politicians are more likely than low-ability politicians to implement the policy early in their term.³

More generally, outside the formal electoral accountability literature, several scholars have studied policy timing driven by policy-motivated actors, examining strategic delay in judicial decisions, executive nominations, and regulation (Krehbiel, 2021; Ostrander, 2016; Binder and Maltzman, 2002; Thrower, 2018; Potter, 2017). In these accounts, timing reflects ideological conflict or bargaining strategies. We focus on how electoral incentives where politicians are concerned with appearing competent.

In our focus on the process of policy development, we are similar to Shaver (2025); Judd (2017); Hitt, Volden and Wiseman (2017), all of which ignore dynamic considerations, instead focusing on a static decision to implement a policy after learning about its quality. In our incorporation of a direct benefit, which the politician enjoys if he implements the policy before the end of his term, our model is most similar to Shaver (2025), which focuses on how an incumbent politician’s ideological preferences shape what a voter infers from his decision to implement a reform or not. Like Shaver (2025), we show that as the politician enjoys a larger direct benefit from policy implementation, the voter updates less positively about his ability given he implements it.

Finally, our model is closely related to formal work on reputation and dynamic information disclosure (e.g., Acharya, DeMarzo and Kremer, 2011; Guttman, Kremer and Skrzypacz, 2014; Zhou, 2024). In Gratton, Holden and Kolotilin (2018), the paper most similar to ours, a sender is privately informed of her type and cares about her reputation at the end of the

³A related set of papers study a politician’s decision to flip-flop or remove a policy he implemented earlier in his term (e.g., Majumdar and Mukand, 2004; Andreottola, 2021). While these models are dynamic, the incumbent politician does not choose when to act during his term. Instead, in the first period, he chooses whether to implement a policy. Then, in the second, he chooses whether to reverse his initial decision.

game. During the game, she privately learns whether and when the opportunity to begin a public flow of information about her type arrives; once it does, she chooses whether and when to exercise the option. Our models differ in two ways. First, in Gratton, Holden and Kolotilin (2018), neither type can act (begin revealing information) until the opportunity arrives. In ours, both types can act from the outset, but a high-type politician cannot implement a well-designed policy until one arrives. Second, in our model, the politician receives a direct payoff from policy implementation, whereas in Gratton, Holden and Kolotilin (2018), the sender only cares about her reputation. Despite these *ex ante* differences, *ex post*, the models yield qualitatively similar equilibrium structures as long as the direct benefit from policy implementation is not too large: once a well-designed policy or the arm arrives, the high type acts, and the low type randomizes over each period.⁴ The key to this is the fact that although the high-type has the ability to act before developing a well-designed policy, he strictly prefers to delay when he might be able to develop one in the future. Thus, in equilibrium, it is equivalent to his not having the opportunity to act.

3 Model

We study a model in which an incumbent *politician* (“he”) decides whether and when to implement a particular policy over a two-period term that leads up to an election in which a *voter* (“she”) chooses whether to reelect him. The politician chooses whether to implement the policy in each period, $t \in \{1, 2\}$, with implementation ($a_t = 1$) being a one-time, irreversible action. If the policy is implemented in the first period, policymaking ends; otherwise, the politician may implement the policy in the second period. An unmodeled election occurs after the second period.

⁴Gratton, Holden and Kolotilin (2018) focus on the essentially unique divine equilibrium, but other equilibria exist with different structures.

Politician Types The politician's type is his private information: $\theta \in \Theta = \{H, L\}$, where H denotes a *high-ability* type and L a *low-ability* type. The prior probability that the politician is the high type is $p \equiv \Pr(\theta = H) \in (0, 1)$.

Policy Quality and Development In each period, the policy available to the politician is either *well designed* or *badly designed*. At the start of period t , the politician privately observes the policy's quality, $q_t \in \{0, 1\}$, where $q_t = 1$ denotes that the policy is well designed.

A high-ability politician develops a well-designed policy with probability $\lambda \in (0, 1)$ at the start of each period. Once developed, a well-designed policy remains available until implemented. A low-ability politician never develops a well-designed policy.

Policy Success Once implemented, a policy may generate visible success. A well-designed policy implemented in period t succeeds in that period, $s_t = 1$, with probability $\rho \in (0, 1)$. Otherwise, $s_t = 0$. If implemented in period 1 and not immediately successful, a well-designed policy may succeed in period 2 with the same probability. A badly designed policy never generates visible success.

Voter Information and Beliefs The voter observes the politician's actions, (a_1, a_2) . The politician may implement early $(a_1, a_2) = (1, \emptyset)$, implement late, $(0, 1)$, or not implement at all, $(0, 0)$. If the policy is implemented, the voter also observes whether it succeeds before the election. Because only high-ability politicians can generate success, the voter updates in the same way following success in either period. We denote success by $\hat{s} = 1$, no success by $\hat{s} = 0$, and no implementation by $\hat{s} = \emptyset$. An information set for the voter is therefore

$$\sigma = (a_1, a_2, \hat{s}) \in \Sigma = \{(1, \emptyset, 0), (1, \emptyset, 1), (0, 1, 0), (0, 1, 1), (0, 0, \emptyset)\}.$$

At each information set, the voter forms a posterior belief about the politician's type, denoted $\mu(\sigma) \equiv \Pr(\theta = H \mid \sigma)$.

Payoffs The politician values both his reputation for competence and a private direct benefit from implementing the policy. His utility is

$$U_P = \mu(\sigma) + b \cdot \mathbb{1}\{(a_1, a_2) \neq (0, 0)\},$$

where $b \geq 0$. The reputation term reflects the politician's probability of reelection in the unmodeled election following the policymaking stage.⁵ The direct benefit, b , captures potential benefits that come from control over distributive resources released by policy implementation itself, independent of the policy's quality.

The voter prefers a well-designed policy to the status quo and the status quo to a badly designed policy. Her utility is

$$U_V = \begin{cases} 1 & \text{if } a_1 = q_1 = 1 \text{ or } a_1 = 0, a_2 = q_2 = 1, \\ 0 & \text{if } a_1 = a_2 = 0, \\ -\zeta & \text{otherwise,} \end{cases}$$

where $\zeta > 0$.

Sequence of Play To summarize, the sequence of play is as follows:

0. Nature draws the politician's type $\theta \in \{H, L\}$ and a policy-quality sequence $(q_1, q_2) \in \{(1, 1), (0, 1), (0, 0)\}$.
1. The politician observes (θ, q_1) and chooses $a_1 \in \{0, 1\}$. If $a_1 = 0$, the game proceeds to (3).
2. If $a_1 = 1$ and $q_1 = 1$, the policy succeeds with probability ρ ; otherwise, $s_1 = 0$ and the game proceeds to (4).

⁵This payoff structure can be microfounded in a model with an explicit election in which the voter reelects the politician if and only if $\mu(\sigma)$ exceeds the challenger's expected ability, which is drawn from a uniform distribution on the unit interval after policymaking concludes.

3. The politician observes q_2 and chooses $a_2 \in \{0, 1\}$. If $a_2 = 0$, the game proceeds to (5).
4. If a well-designed policy has been implemented and has not yet succeeded, it succeeds with probability ρ ; otherwise, $s_2 = 0$.
5. The voter observes $(a_1, a_2, \hat{s})$, forms beliefs, and payoffs are realized.

Strategies and Solution Concept A strategy specifies, for each non-terminal history, the probability with which the politician implements the policy given his type and available policy quality. Formally:

- (i.) In $t = 1$, for each $(\theta, q_1) \in \{(H, 1), (H, 0), (L, 0)\}$, a distribution over $a_1 \in \{0, 1\}$.
- (ii.) In $t = 2$, for each $(\theta, q_1, q_2) \in \{(H, 1, 1), (H, 0, 1), (H, 0, 0), (L, 0, 0)\}$, a distribution over $a_2 \in \{0, 1\}$.

The solution concept is perfect Bayesian equilibrium satisfying D1 (Banks and Sobel, 1987).

4 Discussion of the Model

Before analyzing the model, we discuss its main assumptions and their substantive interpretations. The model captures, in tractable form, the policymaking environment described in the introduction. Politicians have discretion over when to act, effective policy takes time to develop, politicians differ in their ability to do so, and policy effects take time to materialize.

4.1 Politician's Type and λ

The parameter λ governs the development or availability of a well-designed policy. One interpretation is that politicians differ directly in their capacity to generate effective policy. Some executives possess the leadership, managerial skill, staff quality, access to expertise, and coalition-building ability needed to develop a well-designed policy with some regularity,

while others do not. Under this interpretation, high-ability politicians generate a meaningful stream of viable opportunities for action, whereas low-ability politicians do not.

A second interpretation is that policy opportunities arise stochastically, but politicians differ in their ability to exploit them. Crises, legal decisions, shifts in public opinion, or changes in coalition structure may create openings for action. Whether those openings become effective policy depends on political and managerial competence. High-ability politicians are better able to convert such opportunities into viable, well-designed policy.

These complementary interpretations capture the same underlying distinction between types. Politicians differ in their ability to produce policy that is viable and likely to work. Our analysis examines how differences in that ability shape politicians' actions and what voters infer from their behavior.

4.2 Policy Quality

In the main text, policy quality is binary: a well-designed policy is either available or it is not. This rules out the possibility that a politician who has developed a policy that is higher quality than the status quo might delay in hopes of developing an even better one. Appendix B considers a more general formulation of the baseline model in which policy quality varies continuously. This generalization yields substantively similar results. In particular, core insights remain: the low type attempts to mimic an unlucky high type, the low type delays implementation relative to the high type, and policies implemented early are higher quality on average. The binary case is used in the main text because it simplifies the exposition and isolates the electoral incentives of interest, abstracting from an additional motive to wait that would arise even in the absence of electoral considerations.

4.3 Voter Information

A central informational asymmetry in the model is that voters do not observe whether a well-designed policy is available to the politician—or, under the second interpretation of λ ,

whether a policy window has emerged. Many determinants of feasibility, including bureaucratic or legal constraints, preferences of veto players, and the coherence of the technical details of policy, are internal to the policymaking process and not directly observable to the public. As a result, voters cannot reliably infer whether inaction reflects the absence of a viable opportunity or a failure of political competence until a policy is implemented and its consequences begin to unfold.

4.4 Visibility of Policy Success

Another key feature of the model is that until an implemented policy succeeds, voters cannot distinguish between a policy that is badly designed and one that is well designed but slow to show results. The parameter ρ captures the speed with which a well-designed policy becomes visibly successful. Substantively, it represents the expected time between a policy action and its observable effects. This varies naturally across policy areas. In some policy domains, such as macroeconomic stabilization or emergency public-health interventions, effective policy may produce visible evidence relatively quickly. In others, such as environmental policy or disaster preparedness, well-designed policy may take substantial time to show whether it is working.

We assume that badly-designed policies are observationally equivalent to well-designed policies that have not yet succeeded. Allowing badly-designed policies to generate distinctively negative outcomes would weaken the incentive to pursue them; abstracting from this force yields a cleaner characterization of the incentives at the center of the analysis. Substantively, this assumption captures settings in which unsuccessful policy change appears to the public less like active harm than the continuation of a bad status quo—for example, anti-inflationary measures that fail to stabilize prices.

4.5 Politician Payoff

The politician's utility reflects two concerns: maintaining a reputation for competence and obtaining a direct benefit from policy implementation. The reputational component captures the electoral incentive to be perceived as a high-ability type. The direct benefit, captured by b , represents a private reward from implementation itself that is orthogonal to whether the policy is effective. A useful interpretation of b is as the private benefit a politician receives when implementing the policy allows contracts or other material benefits to be directed to himself or to allied interests. Consider a municipal politician facing pressure to address a traffic problem. Determining which project would solve the problem effectively may take time and may require evaluating several alternatives. But once the politician moves forward with some project, he may be able to direct the associated contract to a firm in which he, his associates, or an allied group has a stake. In that case, implementation generates a private benefit even if the chosen project is not the one most likely to solve the traffic problem.

The parameter b is publicly known. Voters know when the political environment gives politicians especially strong private rewards from moving forward with policy, and they incorporate that into their interpretation of action and inaction. Higher values of b may reflect either politician-specific characteristics or features of the policy domain itself. Politicians regarded as corrupt would naturally be associated with higher b , and policy areas such as infrastructure or defense procurement would generally correspond to higher b than domains in which implementation creates fewer opportunities for material private reward.

To isolate electoral incentives and keep the model simple, we assume that politicians do not directly value the quality of policy. In equilibrium, high-ability politicians behave as if they care directly about policy quality, since they implement only when a well-designed policy is available. Low-ability politicians implement badly designed policies in equilibrium to appear competent in response to electoral incentives. Adding a direct payoff from policy quality would not alter the basic strategic logic, but would attenuate the degree to which low-ability politicians implement badly designed policies.

4.6 Voter Payoff

The voter prefers a well-designed policy to the status quo and the status quo to a badly designed policy. At the end of the game, the voter receives the payoff from a well-designed policy regardless of whether that policy generated visible success before the election. What matters electorally is whether success becomes visible before the election, because visible success provides information about policy quality and politician competence. Adding a preference for early success over late success over success realized only after the election would not change the voter's inference about the politician, which is what shapes the politician's electoral incentives in the model.

5 Analysis

5.1 Preliminaries

We begin by refining notation for strategies and beliefs and by stating basic properties that must hold in any equilibrium.

There are five possible information sets for the voter. Because only the high type can generate visible success ($\hat{s} = 1$), success fully reveals the politician's type:

$$\Pr(\theta = H \mid a_1 = 1, \hat{s} = 1) = \Pr(\theta = H \mid a_1 = 0, a_2 = 1, \hat{s} = 1) = 1.$$

Accordingly, we focus on the remaining three *no-success* information sets, indexed by $i \in \{1, 2, \phi\}$, where

$$i = \begin{cases} 1 & \text{if } a_1 = 1 \text{ and } \hat{s} = 0 \quad (\text{early implementation, no success}), \\ 2 & \text{if } a_1 = 0, a_2 = 1 \text{ and } \hat{s} = 0 \quad (\text{late implementation, no success}), \\ \phi & \text{if } a_1 = a_2 = 0 \quad (\text{no implementation}). \end{cases}$$

We write $\mu_i \equiv \Pr(\theta = H \mid i)$ for the voter's posterior belief at each $i \in \{1, 2, \phi\}$.

The politician's strategy specifies whether to implement the policy in each period, conditional on his type and, for the high type, the quality of the available policy. The low type's strategy can be summarized by the probabilities

$$\gamma_1 \equiv \Pr(a_1 = 1 \mid \theta = L), \quad \gamma_2 \equiv \Pr(a_1 = 0, a_2 = 1 \mid \theta = L), \quad \gamma_\phi \equiv 1 - \gamma_1 - \gamma_2,$$

where γ_ϕ denotes the probability of no implementation.⁶

Lemma 1 characterizes properties that must hold in any equilibrium.

Lemma 1. *In any equilibrium:*

1. *The low type implements the policy with positive probability in each period: $\gamma_1 \in (0, 1)$ and $\gamma_2 \in (0, 1)$.*
2. *The high type implements the policy in $t = 2$ if it is well designed and in $t = 1$ if and only if it is well-designed.*

To see why the low type must mix between implementation and non-implementation in both periods, suppose instead that $\gamma_1 = 0$. Then any early implementation would be off the equilibrium path since if the high type implemented early with positive probability, early implementation would fully reveal the incumbent to be a high type, even if policy does not yield visible success. But because visible success always fully reveals the high type, and because D1 assigns deviations to the type with the strongest incentive to deviate, early implementation without success would be attributed to a high type with a well-designed policy. Anticipating this, the low type would strictly prefer to deviate by implementing early, contradicting $\gamma_1 = 0$. A symmetric argument rules out $\gamma_2 = 0$.⁷

⁶Formally, (γ_1, γ_2) is sufficient to describe the low type's strategy if $\gamma_1 < 1$, since $\Pr(a_2 = 1 \mid a_1 = 0, \theta = L) = \gamma_2 / (1 - \gamma_1)$. Lemma 1 establishes that $\gamma_1 = 1$ cannot occur in equilibrium.

⁷Lemma 1 does not rule out $\gamma_1 + \gamma_2 = 1$: while the low type must mix across periods, no implementation need not occur with positive probability.

Given that the low type mixes between early and late implementation, he must be indifferent across these actions and weakly prefer implementation to inaction: $b + \mu_1 = b + \mu_2 \geq \mu_\phi$. It follows that the voter assigns the same posterior belief following unsuccessful early and late implementation, so $\mu_1 = \mu_2$. Given this belief structure, a high type strictly prefers to implement a well-designed policy immediately, in order to maximize the probability of visible success, and strictly prefers to delay when the policy is badly designed, since he may develop a well-designed policy in the next period.

5.2 Welfare-Maximizing Equilibrium

The first part of Lemma 1 pins down the low type's equilibrium behavior given any strategy of the high type consistent with the second part. However, multiple equilibria may arise because the high type may, with some probability, implement the policy in the second period even when it is badly designed. In particular, there always exists an equilibrium in which the high type implements the policy with probability one in $t = 2$ after failing to develop a well-designed policy in either period.⁸

When the direct benefit from implementation is sufficiently large, this equilibrium is unique: if b is large enough, the high type strictly prefers implementing the policy—even at the cost of certain electoral defeat—to refraining and securing reelection. To focus on environments in which electoral incentives are strong enough to deter low-quality implementation by the high type, we restrict attention to cases in which the direct benefit is small enough that the high type refrains from implementing a badly designed policy. Assumption 1 formalizes this restriction.⁹

Assumption 1. *The direct benefit from policy implementation is sufficiently small that the*

⁸In such equilibria, the low type implements as well in order to mimic the high type, mixing between early and late implementation so that $\mu_1 = \mu_2$. As long as voter beliefs following unexpected non-implementation are sufficiently unfavorable, neither type has an incentive to deviate from implementing in $t = 2$; such beliefs satisfy D1. We formally characterize this equilibrium in the Appendix.

⁹The condition in Assumption 1 is derived in the Appendix.

politician's electoral incentives deter the high type from implementing a badly designed policy:

$$b < \bar{b} \equiv \frac{1 - p}{(1 - p) + p\lambda(1 - \rho)(2 - \lambda - \rho)}.$$

Under Assumption 1, a continuum of equilibria exist that differ in the probability with which the high type implements the policy in $t = 2$ after failing to develop a well-designed version. Denote this probability by ω . Among these equilibria, the voter's welfare—defined as her *ex ante* expected equilibrium payoff,

$$V^* \equiv p(1 - (1 - \lambda)^2) - \zeta(1 - p)(\gamma_1 + \gamma_2 + \omega), \quad (1)$$

—is maximized when $\omega = 0$. We refer to this equilibrium as the *welfare-maximizing equilibrium*.

Lemma 2. *There exists a unique welfare-maximizing equilibrium. In this equilibrium, the high type implements the policy in the second period if and only if it is well designed, $\omega = 0$, and the low type chooses no implementation with positive probability, $\gamma_\phi \in (0, 1)$.*

The intuition is straightforward. As the high type becomes more likely to implement a badly designed policy in $t = 2$, he implements the policy more often overall. The low type must then increase the probability he implements the policy as well in order to remain indistinguishable, increasing the frequency with which badly-designed policies are implemented in equilibrium. Voter welfare therefore declines both because the high type sometimes implements a badly designed policy and because the low type mimics this behavior. Setting $\omega = 0$ minimizes these distortions and uniquely maximizes voter welfare. For the remainder of the paper, we focus on this welfare-maximizing equilibrium.

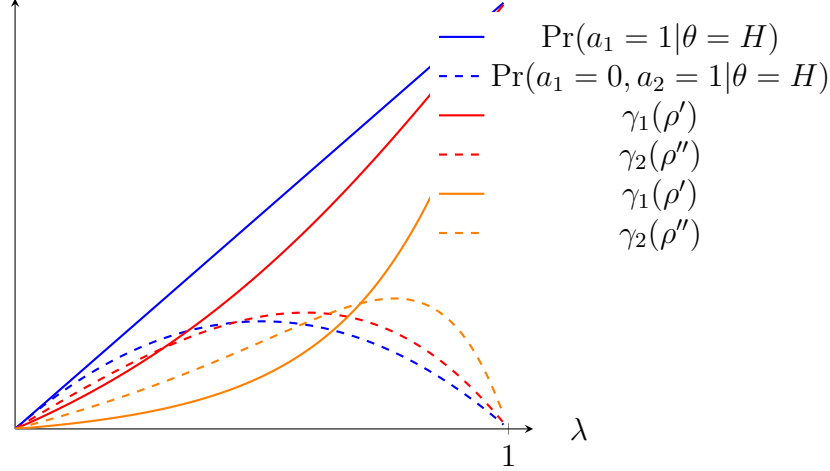


Figure 1: Probability of early and late implementation by politician type when $b = 0$; $\rho' = \frac{3}{10}$ and $\rho'' = \frac{7}{10}$

5.3 Equilibrium Characterization

We now fully characterize the equilibrium. To build intuition for the general case, $b \in [0, \bar{b})$, we begin with a benchmark case $b = 0$, in which the politician values only his reputation. This case is depicted in Figure 1.

Remark 1. *If $b = 0$, the low type's equilibrium strategy is*

$$\gamma_1 = \frac{\lambda(1-\rho)^2}{\mathcal{H}}, \quad \gamma_2 = \frac{(1-\lambda)\lambda(1-\rho)}{\mathcal{H}},$$

and the voter's posterior beliefs satisfy

$$\mu_i = \frac{p}{p + \frac{1-p}{\mathcal{H}}}$$

for each $i \in \{1, 2, \phi\}$, where $\mathcal{H} = \Pr(i \in \{1, 2, \phi\} \mid \theta = H)$.

The low type's objective is to mimic the behavior of the high type subject to the constraint that, unlike the high type, he can never generate visible policy success. We therefore begin by characterizing the distribution of no-success information sets generated by the high type.

By Lemmas 1 and 2, the high type implements the policy in period 1 with probability λ ,

in period 2 with probability $(1 - \lambda)\lambda$, and not at all with probability $(1 - \lambda)^2$. Conditional on these actions failing to generate visible success, the induced distribution over the no-success information sets is

$$\begin{aligned} h_1 &= \lambda(1 - \rho)^2 \quad (\text{early implementation, no success}), \\ h_2 &= (1 - \lambda)\lambda(1 - \rho) \quad (\text{late implementation, no success}), \\ h_\phi &= (1 - \lambda)^2 \quad (\text{no implementation}). \end{aligned}$$

Let $\mathcal{H} \equiv h_1 + h_2 + h_\phi < 1$ denote the probability that a high type generates a no-success information set.

The low type mixes over early implementation, late implementation, and no implementation with probabilities $(\gamma_1, \gamma_2, \gamma_\phi)$. By Bayes' rule, the voter's posterior belief following a no-success information set i is

$$\mu_i = \frac{p h_i}{p h_i + (1 - p)\gamma_i}.$$

In equilibrium, the low type chooses $(\gamma_1, \gamma_2, \gamma_\phi)$ so that these posteriors coincide across all no-success information sets,

$$\mu_1 = \mu_2 = \mu_\phi, \tag{2}$$

which implies proportional likelihoods,

$$\frac{h_1}{\gamma_1} = \frac{h_2}{\gamma_2} = \frac{h_\phi}{\gamma_\phi}.$$

Thus, the low type's equilibrium strategy uniquely satisfies

$$\gamma_1 = \frac{h_1}{\mathcal{H}}, \quad \gamma_2 = \frac{h_2}{\mathcal{H}}, \quad \gamma_\phi = \frac{h_\phi}{\mathcal{H}},$$

yielding the expressions in Remark 1.

Notably, the low type does not perfectly imitate the high type's implementation frequencies. Perfect imitation would imply $\mu_\phi = p$, since non-implementation would be equally likely under both types. But because policies implemented by the high type may succeed while those implemented by the low type never do, unsuccessful implementation is worse news for the voter than inaction, implying $\mu_1 < \mu_\phi$. Perfect imitation would therefore make early implementation strictly unattractive for the low type.

Instead, the low type behaves like an “unlucky” high type—one who either implements a policy that fails to show visible success or never develops a well-designed policy. By mixing across early implementation, late implementation, and inaction in the appropriate proportions, he reproduces the distribution of no-success information sets generated by the high type. When $b = 0$, this requires the low type to implement the policy less frequently than the high type.

We now extend this characterization to the general case $b \in [0, \bar{b})$. When $b > 0$, the low type trades off reputational concerns against the direct benefit from implementation, so the equilibrium indifference condition becomes

$$\mu_1 + b = \mu_2 + b = \mu_\phi. \quad (3)$$

Let $\mu_R = \mu_1 = \mu_2$ denote the posterior following unsuccessful implementation.

Proposition 1. *The low type's equilibrium strategy is*

$$\gamma_1 = k(b)h_1, \quad \gamma_2 = k(b)h_2, \quad \gamma_\phi = 1 - k(b)(h_1 + h_2),$$

where $k(b) : [0, \bar{b}] \rightarrow [\mathcal{H}^{-1}, (h_1 + h_2)^{-1}]$ is strictly increasing. The high type implements the policy in period t if and only if it is well designed:

$$\Pr(a_t = 1 \mid \theta = H, q_t = 1) = 1, \quad \Pr(a_t = 1 \mid \theta = H, q_t = 0) = 0.$$

The voter's equilibrium beliefs are

$$\mu_R = \frac{p}{p + (1-p)k(b)}, \quad \mu_\phi = \frac{ph_\phi}{ph_\phi + (1-p)(1-k(b)(h_1 + h_2))}.$$

As b increases, the low type implements the policy more frequently, but allocates this additional implementation proportionally across early and late periods so as to preserve the relative likelihoods that equalize voter beliefs across no-success information sets. This adjustment is summarized by the function $k(b)$.¹⁰

5.4 Comparative Statics

Having characterized equilibrium behavior, we now study how variation in λ , ρ , and b affects the politician's behavior and equilibrium outcomes.

5.4.1 Probability of Policy Implementation

We first examine how the probability of policy implementation varies across environments, both overall and conditional on the politician's type.

Proposition 2. *In equilibrium, the probability that the low type implements the policy, $\gamma_1 + \gamma_2$, decreases in ρ and increases in λ and b .*

Since the high type's implementation probability is pinned down directly by the rate at which high-quality policy arrives, the main behavioral comparative statics concern the low type's implementation decision.¹¹ The implication for the unconditional probability of implementation then follows immediately.

¹⁰We characterize $k(b)$ formally in the Appendix.

¹¹The probability that the high type implements the policy, $1 - h_\phi$, is increasing in λ and constant in ρ and b .

Corollary 1. *The unconditional probability that the policy is implemented,*

$$p(\gamma_1 + \gamma_2) + (1 - p)(1 - h_\phi),$$

decreases in ρ and increases in λ and b .

The intuition behind Proposition 2 and Corollary 1 follows from the low type's incentive to mimic the high type across no-success information sets.

An increase in ρ raises the probability that a well-designed policy produces visible success before the election. Although this does not affect the high type's implementation decision, it reduces the likelihood that his actions lead to unsuccessful implementation, shrinking $h_1 + h_2$ relative to h_ϕ . Unsuccessful implementation therefore becomes stronger evidence of low ability, inducing the low type to implement the policy less often.

An increase in λ makes the high type more likely to develop and implement a well-designed policy, shifting probability mass away from no implementation and toward unsuccessful implementation. Because implementation—conditional on no success—becomes more characteristic of the high type, the low type must increase his own frequency of implementation to remain indistinguishable.

Finally, an increase in b raises the direct payoff from implementation for the low type. The equilibrium condition $b + \mu_R = \mu_\phi$ implies that a larger b makes implementation more attractive, leading the low type to implement the policy more frequently. Since the high type's strategy is unaffected by b , this adjustment occurs through changes in voter beliefs: μ_R falls and μ_ϕ rises until indifference is restored.

5.4.2 Timing of Policy Implementation

We now examine how λ , ρ , and b affect the *timing* of policy implementation. We focus on the probability of early versus late implementation *conditional on implementation occurring*, abstracting from variation in the overall frequency of implementation.

Proposition 3. *Conditional on the politician implementing the policy during his term:*

1. *The odds that the high type implements early are increasing in λ and constant in ρ and b .*
2. *The odds that the low type implements early are increasing in λ , decreasing in ρ , and constant in b .*
3. *The odds that the policy is implemented early, unconditional on type, are decreasing in b .*

For the high type, the odds of early relative to late implementation are

$$\Omega_H \equiv \frac{\Pr(\text{early} \mid \text{implementation}, \theta = H)}{\Pr(\text{late} \mid \text{implementation}, \theta = H)} = \frac{1}{1 - \lambda}.$$

Because the high type implements if and only if the policy is well designed, his odds of implementing early depend only on λ . As λ increases, he is more likely to develop a well-designed policy in the first period, raising the likelihood of early implementation relative to late implementation.

For the low type, the odds of early relative to late implementation are

$$\Omega_L \equiv \frac{\Pr(\text{early} \mid \text{implementation}, \theta = L)}{\Pr(\text{late} \mid \text{implementation}, \theta = L)} = \frac{h_1}{h_2} = \frac{1 - \rho}{1 - \lambda}.$$

The low type's odds of implementing early therefore depend on both λ and ρ . As λ increases and the high type becomes more likely to implement early, the low type shifts toward earlier implementation to remain indistinguishable. By contrast, as ρ increases, unsuccessful early implementation becomes relatively less likely than unsuccessful late implementation, since early policies have more time to succeed. The low type therefore shifts probability mass toward late implementation to preserve the distribution of no-success information sets. Although an increase in b raises the low type's overall frequency of implementation, it does

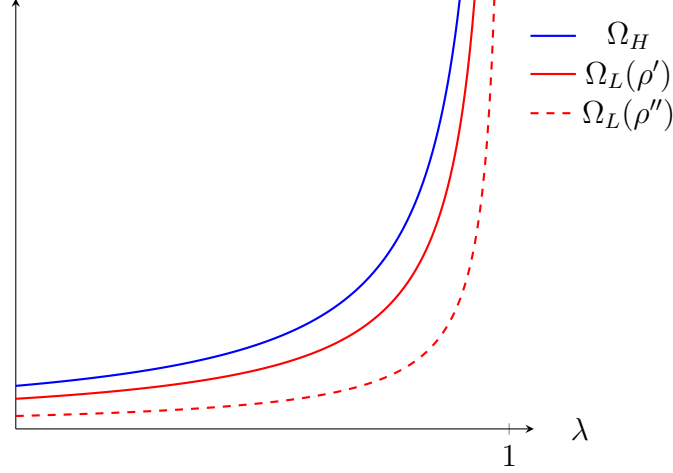


Figure 2: Odds of early relative to late implementation for the low and high types; $\rho' = \frac{3}{10}$ and $\rho'' = \frac{7}{10}$

not affect his odds of early implementation: the additional implementation probability is distributed proportionally across both periods to maintain observational equivalence.

Figure 2 summarizes these patterns. In equilibrium, the odds of early implementation are always lower for the low type than for the high type, since

$$\Omega_L = (1 - \rho)\Omega_H.$$

Were this not the case, the voter would correctly interpret a lack of visible success after early policy implementation as stronger evidence of low ability than a late policy that similarly fails to visibly succeed. Low types therefore implement the policy later on average.

Corollary 2. *The odds of early implementation are always lower for the low type than for the high type: $\Omega_H > \Omega_L$ for all λ, ρ, p , and b .*

The overall odds of early implementation, unconditional on type, are

$$\Omega \equiv \frac{\Pr(\text{early} \mid \text{implementation})}{\Pr(\text{late} \mid \text{implementation})} = \frac{p\lambda + (1-p)k(b)h_1}{p(1-\lambda)\lambda + (1-p)k(b)h_2}.$$

Although b does not affect the odds of early implementation conditional on type, it does

affect the overall odds of early implementation. In particular, as b increases, the share of policy implementation carried out by low types, who are systematically more likely to implement late, also increases. Hence, Ω increases. As a result, stronger direct incentives for policy implementation shift the aggregate distribution of implementation towards the second period. This yields a potentially counterintuitive empirical implication: holding other factors fixed, policies associated with larger direct benefits to politicians should be less likely to be implemented early in the politician's term.

5.4.3 Policy Quality Conditional on Timing

We now examine how the quality of an implemented policy varies with when it is implemented. Specifically, we study the probability that a policy is well designed, conditional on whether it is implemented early or late in the politician's term. Using equilibrium strategies, these probabilities are

$$\Pr(\text{well-designed} \mid a_1 = 1) = \frac{p\lambda}{p\lambda + (1-p)\gamma_1} = \frac{p}{p + \left(\frac{1-p}{k(b)\mathcal{H}}\right)(1-\rho)^2},$$

$$\Pr(\text{well-designed} \mid a_1 = 0, a_2 = 1) = \frac{p(1-\lambda)\lambda}{p(1-\lambda)\lambda + (1-p)\gamma_2} = \frac{p}{p + \left(\frac{1-p}{k(b)\mathcal{H}}\right)(1-\rho)}.$$

As before, we summarize differences across periods using odds, defined as the ratio of the probability that a policy is well designed to the probability that it is badly designed conditional on when it is implemented:

$$\Omega_1 = \frac{p\lambda}{(1-p)\gamma_1}, \quad \Omega_2 = \frac{p(1-\lambda)\lambda}{(1-p)\gamma_2}.$$

The odds ratio Ω_1/Ω_2 captures how much more likely the policy is to be well designed when implemented early in the politician's term relative to late.

Proposition 4.

1. *Policies implemented early are always more likely to be well designed than policies implemented late:*

$$\Pr(\text{well-designed} \mid a_1 = 1) > \Pr(\text{well-designed} \mid a_1 = 0, a_2 = 1) \quad \forall \lambda, \rho, p, b.$$

2. *The odds that policy implemented early is well designed relative to policy implemented late are increasing in ρ and constant in λ , p , and b :*

$$\frac{\Omega_1}{\Omega_2} = \frac{1}{1 - \rho}.$$

The intuition for the first result follows from Corollary 2. When the low type implements policy, he does so early less frequently compared to the high type in order to mimic an unlucky high type. Thus a larger share of early policy implementations are carried out by the high type, meaning policies implemented early are more likely to be well designed.

The second result describes how the quality gap between policies implemented early versus late varies. As ρ increases, well-designed policies that have been implemented are revealed as such more often, and this advantage is disproportionately concentrated among early implementations, which have more time to succeed. The low type responds by shifting further away from early implementation, widening the quality gap between early and late policies. In contrast, changes in λ , p , or b affect the frequency of policy implementation but not its relative quality by timing.

These results yield clear empirical implications. Policies implemented earlier in a politician's term should, holding ρ fixed, be well designed more often than those implemented later. Moreover, this quality gap should be small in domains where policy effects are slow to materialize and grow larger as outcomes become more quickly observable. Because the odds ratio Ω_1/Ω_2 is independent of λ , p , and b , the model predicts this timing-quality relationship to persist across environments that differ in politician quality, direct implementation

incentives, and the frequency of well-designed policy opportunities.

5.4.4 Voter Beliefs

We now examine the voter's beliefs about the incumbent's ability. We begin by studying how the voter's interim and terminal beliefs are ordered and how variation in the model's parameters shape this ordering. We then study how variation in the model's parameters influence beliefs formed at the end of the term when success has not materialized.

Let $\hat{\mu}_t$ denote the voter's belief that the incumbent is the high type at the moment he implements the policy in period t :

$$\hat{\mu}_1 \equiv \Pr(\theta = H \mid a_1 = 1), \quad \hat{\mu}_2 \equiv \Pr(\theta = H \mid a_1 = 0, a_2 = 1),$$

and let

$$\hat{\mu}_\phi \equiv \Pr(\theta = H \mid a_1 = 0)$$

denote the voter's belief following delay in the first period.

Proposition 5.

1. *If b is sufficiently low and λ is sufficiently low, then*

$$\hat{\mu}_1 > \hat{\mu}_2 > p > \hat{\mu}_\phi > \mu_\phi \geq \mu_R.$$

2. *If b is sufficiently low and λ is sufficiently high, then*

$$\hat{\mu}_1 > p > \hat{\mu}_\phi > \hat{\mu}_2 > \mu_\phi \geq \mu_R.$$

3. *If b is sufficiently high and λ is sufficiently low, then*

$$\mu_\phi > \hat{\mu}_\phi > p > \hat{\mu}_1 > \hat{\mu}_2 > \mu_R.$$

4. If b is sufficiently high and λ is sufficiently high, then

$$\mu_\phi > \hat{\mu}_1 > p > \hat{\mu}_\phi > \hat{\mu}_2 > \mu_R.$$

Despite this heterogeneity in different regions of the parameter space, the voter's interim beliefs when the politician implements policy early versus late satisfy a consistent ordering.

Corollary 3. *The voter's interim belief is always higher following early implementation than following late implementation: $\hat{\mu}_1 > \hat{\mu}_2$.*

Because the low type delays implementation to imitate an unlucky high type, late implementation is more likely to be undertaken by a low type. Early implementation therefore provides a stronger interim signal of competence. However, as time progresses without visible success, the voter's posterior declines such that when an unsuccessful policy was implemented is no longer informative on election day.

Relatedly, the voter's beliefs are always most pessimistic when the politician's term ends with an implemented policy that has failed to produce visible success.

Corollary 4. *The voter's posterior belief is always lowest following implementation without visible success: $\mu_R < \min\{p, \hat{\mu}_2, \hat{\mu}_\phi\}$ and $\mu_R \leq \mu_\phi$ with strict inequality if $b > 0$.*

This follows because, when success fails to materialize by the end of the term, the possibility of future success has been exhausted. When $b > 0$, the low type's equilibrium indifference condition $b + \mu_R = \mu_\phi$ further implies that non-implementation is strictly more favorable than unsuccessful implementation.

Whether implementation initially improves or harms the incumbent's reputation depends on b and λ . When b is low, the low type implements less frequently than the high type early in the term, so early implementation improves the incumbent's interim reputation. When λ is also low, the high type delays sufficiently often that late implementation may also provide a temporary reputational boost. When b is high, by contrast, the low type implements more

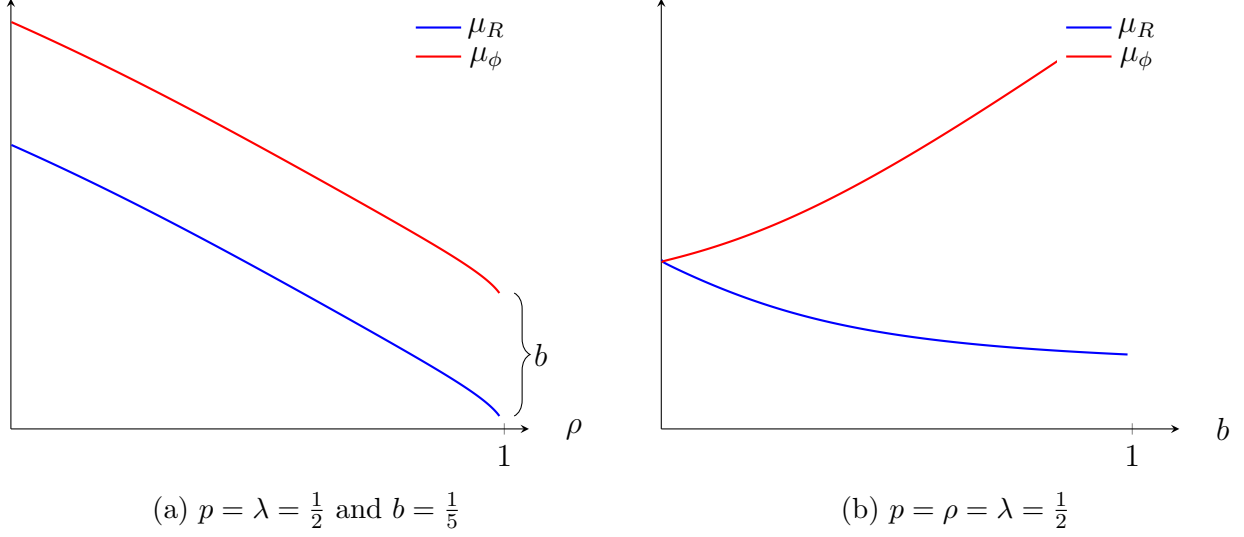


Figure 3: Voter's posterior as a function of ρ and b

frequently than the high type, particularly in the second period, making late implementation initially harmful to the incumbent's reputation and non-implementation beneficial. Early inaction can also send an initial signal of competence in such settings if high types are sufficiently unlikely to develop a well-designed policy early in their term, i.e., when λ is sufficiently low.

We now turn to comparative statics of the voter's terminal beliefs.

Proposition 6.

1. μ_R is decreasing in ρ , decreasing in b , decreasing in λ when b is small, and non-monotonic in λ when b is large.
2. μ_ϕ is decreasing in ρ , increasing in b , decreasing in λ when b is small, and non-monotonic in λ when b is large.

An increase in ρ makes the absence of success increasingly inconsistent with high ability, lowering both μ_R and μ_ϕ . This is illustrated in the left panel of Figure 3. The gap between these posteriors equals b , reflecting the equilibrium condition $b + \mu_R = \mu_\phi$.

An increase in b shifts beliefs in opposite directions, as shown in the right panel of Figure 3. Because a higher b induces low types to implement more frequently, unsuccessful im-

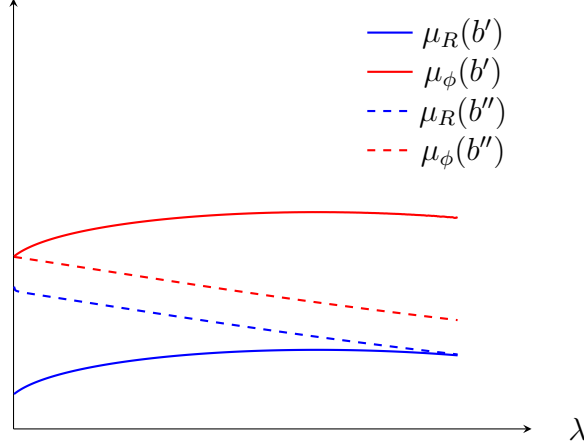


Figure 4: Voter's posterior as a function of λ ; $p = \rho = \frac{1}{2}$, $b' = \frac{2}{5}$, and $b'' = \frac{1}{10}$

plementation becomes more indicative of low ability, reducing μ_R , while non-implementation becomes rarer among low types, raising μ_ϕ .¹²

The effect of λ is more nuanced. When b is small, increases in λ monotonically reduce voter posteriors following non-success: as high types become more likely to develop viable policies, the absence of visible success becomes stronger evidence of incompetence. When b is large, the effect of λ is non-monotonic. For very low λ , policy implementation is rare for both types, and unsuccessful implementation is overwhelmingly attributable to a low type pursuing the direct benefit. As λ increases from this baseline, the likelihood that non-success reflects an unlucky high type initially rises, since the effect of λ on the high type's overall probability of developing a well-designed policy, $1 - (1 - \lambda)^2$, is sharpest when λ is low. Beyond a threshold, however, the high type becomes sufficiently likely to develop a well-designed policy that further increases in λ only marginally affect his behavior, restoring the logic of the low- b case and causing voter posteriors to decline.

¹²This result parallels the observation in Shaver (2025) that a politician's non-electoral incentives shape what is conveyed about his type through policy change. However, in Shaver (2025), this result arises because both types of politician are willing to implement lower-quality policies. In our model, this result emerges from the low type implementing the policy more often.

5.4.5 Welfare

We conclude by examining how the model's parameters affect equilibrium *ex ante* welfare for the voter and politician.

Proposition 7. *The voter's welfare is decreasing in b and increasing in ρ .*

The effect of ρ and b on voter welfare follows directly from Proposition 2. An increase in b or a decrease in ρ raises the frequency with which the low type implements the policy. Because the low type only implements badly-designed policies, these changes reduce voter welfare. Since neither parameter affects the high type's probability of implementing a well-designed policy, this increase in low-quality implementation is the sole channel through which b and ρ affect voter welfare.¹³

We next examine the politician's welfare.

Proposition 8.

1. *If the politician is the low type, his welfare is decreasing in ρ , increasing in b , decreasing in λ when b is sufficiently small, and non-monotonic in λ when b is sufficiently large.*
2. *If the politician is the high type, his welfare is increasing in b . When b is sufficiently small, his welfare is increasing in both λ and ρ .*

Because the low type is indifferent across all no-success histories in equilibrium, his expected payoff is identical in each such information set; his welfare is

$$V_L^* = b + \mu_R = \mu_\phi.$$

The comparative statics follow immediately from Proposition 6. The only subtlety is the comparative static for b since μ_R is decreasing in b while μ_ϕ is increasing in b . However, since (3) must be satisfied in equilibrium, an increase in b has a positive effect overall.

¹³The model does not yield robust comparative statics for voter welfare with respect to λ . As both types implement more frequently when λ rises, the welfare effect depends on parameter values and on how voter utility weights well-designed versus badly-designed policies.

The high type, by contrast, produces visible success with a positive probability, yielding a payoff of one. His equilibrium welfare is

$$\begin{aligned} V_H^* &= b(1 - h_\phi) + (1 - \mathcal{H}) + \mu_\phi h_\phi + \mu_R(h_1 + h_2) \\ &= b + (1 - \mathcal{H}) + \mathcal{H}\mu_R \end{aligned}$$

where the second equality follows from the fact that $\mu_\phi = b + \mu_R$ in equilibrium.

As b rises, the high type directly benefits whenever he develops well-designed policy. When he fails to develop a well-designed policy and thus does not implement the policy, he indirectly obtains an equivalent benefit through $\mu_\phi = b + \mu_R$. Thus the benefit and cost of an increase in b on the high type's welfare can be decomposed into a unit increase in V_H^* and a decrease in μ_R scaled by the probability that he is unlucky. Since $\mu_\phi = b + \mu_\phi$ is increasing in b (Proposition 6), it follows that μ_R declines at a lower rate than b rises. Thus for the high type, the benefit of an increase in b outweighs the cost.

Increases in λ or ρ have offsetting effects on V_H^* . When b is small, as λ or ρ increase, the voter's posterior at a no-success information set decreases. On the other hand, as λ or ρ increase, the probability of reaching a no-success information set decreases. When b is sufficiently small, the latter effect dominates, so the high type benefits from environments with more frequent policy opportunities and faster policy feedback. In addition, higher λ raises the probability that the high type receives the direct benefit from implementation.

Substantively, these results highlight a divergence in preferences over policymaking environments. When reputational concerns dominate (low b), capable politicians benefit from settings with frequent opportunities and visible outcomes, which allow them to demonstrate competence, while less capable politicians prefer environments in which opacity or delay obscures performance differences. High λ and ρ environments expose low types to greater scrutiny, whereas low λ and ρ environments make it more plausible to the voter that he is unlucky rather than incompetent.

6 Conclusion

Policymaking in representative democracy is inherently dynamic. Politicians choose not only what policy to pursue, but also when to act. Effective policy takes time to develop, policy effects take time to materialize, and voters observe only part of that process as it unfolds. This paper studies electoral incentives in that setting. Politicians differ in their ability to develop well-designed policy, may receive direct benefits from implementation, and act before an election while voters infer competence from the timing of action and the presence or absence of visible success. In equilibrium, high-ability politicians implement the policy if and only if a well-designed version is available. Low-ability politicians, by contrast, never develop a well-designed policy and instead randomize across early implementation, late implementation, and inaction in order to mimic an unlucky high-ability politician—either one who has not yet developed a well-designed policy or one who implemented a well-designed policy that failed to produce visible success before the election. As a result, elections discipline high-ability politicians to wait for a well-designed policy before acting, while still leaving room for less competent politicians to implement badly designed policy in order to appear capable.

A central implication of the model is that low-ability politicians delay implementation in equilibrium in the sense that the odds of early implementation are always lower for the low type than for the high type. The reason is that early implementation by a high-ability politician has more time to generate visible success before the election. Low-ability politicians cannot generate success, so they concentrate implementation later and choose mixing probabilities that make the distribution of early and late policies that fail to visibly succeed consistent with that of an unlucky high type. As a result, by election day, unsuccessful early and late policies are observationally equivalent, even though early implementation is initially interpreted more favorably at the moment it occurs. This timing asymmetry yields a sharp prediction about policy quality: policies implemented early in a term are more likely to be well designed than policies implemented later. Moreover, this quality gap is larger

in policy domains where success is expected to materialize quickly and smaller in domains characterized by longer lags.

Our welfare analysis clarifies the consequences of the strategic policymaking dynamics we identify. Voter welfare declines when policy outcomes are less likely to become visible before elections and when politicians face stronger direct incentives to implement the policy, since both forces increase the frequency of badly designed policy. Mapping outcome visibility to real-world policy environments helps clarify the substantive implications of this result. Policies with longer time horizons—where successful outcomes take longer to become observable—can be interpreted as more complex and resource-intensive, such as health care reform, climate change policy, emergency preparedness, or large-scale infrastructure projects, whereas shorter-horizon settings involve more routine or incremental tasks, such as municipal service provision. In these longer-horizon policy areas, policy failures intuitively involve greater resource commitments and higher stakes. In our welfare analysis, however, the voter’s loss from badly designed policy is invariant to outcome visibility. To the extent that failures in longer-horizon policy areas are in fact more costly than failures in shorter-horizon ones, the welfare consequences identified by the model may be more troubling than the formal analysis suggests.

The model also sheds light on voter behavior. In particular, it offers an alternative to accounts that attribute electoral punishment of incumbents who pursue long-horizon or low-visibility policies to shortsightedness of voters. In our model, voters punish incumbents whose policies fail to show success by election day not because they undervalue long-term projects or investments but because uncertainty about the outcomes that these policies may or may not produce creates scope for incompetent politicians to obscure poor performance. Precisely because delay, complexity, and long time horizons are sometimes legitimate features of competent policymaking, they also provide a credible cover for low-ability incumbents. Treating policies that have not demonstrably succeeded with skepticism is therefore a rational response to the informational environment voters face.

The model also yields nuanced implications for how voters respond to political action and inaction over the course of a politician’s term. Early action is always interpreted more favorably than late action at the moment it occurs, yet the reputational consequences of action, delay, and restraint depend systematically on the political environment. Where politicians face strong direct incentives to implement the policy, restraint can improve voter beliefs; where such incentives are weak, delay and inaction are interpreted as signals of low ability. These patterns can help to clarify why public opinion sometimes rewards caution and at other times demands decisiveness.

Finally, our model provides a solid foundation for future research. The politician welfare results in particular suggest extending the analysis to settings in which politicians choose which policy areas to engage. High-ability politicians benefit from opportunity-rich and highly visible domains, while low-ability politicians prefer environments where opportunities are scarce or outcomes are slow to become observable. This suggests incentives for politicians to sort into policy areas based on their ability. Such choices, however, may themselves convey information to voters. A natural direction for future work, then, is to study policy-area selection as an endogenous strategic choice.

References

- Acharya, Viral V., Peter DeMarzo and Ilan Kremer. 2011. “Endogenous Information Flows and the Clustering of Announcements.” *American Economic Review* 101(7):2955–2979.
- Andreottola, Giovanni. 2021. “Flip-flopping and electoral concerns.” *The Journal of Politics* 83(4):1669–1680.
- Ashworth, Scott and Kenneth W. Shotts. 2010. “Does Informative Media Commentary Reduce Politicians’ Incentives to Pander?” *Journal of Public Economics* 94(11–12):838–847.
- Banks, Jeffrey S. and Joel Sobel. 1987. “Equilibrium Selection in Signaling Games.” *Econometrica* 55(3):647–661.
- Bernheim, B Douglas and Aaron L Bodoh-Creed. 2020. “A theory of decisive leadership.” *Games and Economic Behavior* 121:146–168.
- Binder, Sarah A. and Forrest Maltzman. 2002. “Senatorial Delay in Confirming Federal Judges, 1947–1998.” *American Journal of Political Science* 46(1):190–199.
- Canes-Wrone, Brandice, Michael C. Herron and Kenneth W. Shotts. 2001. “Leadership and Pandering: A Theory of Executive Policymaking.” *American Journal of Political Science* 45(3):532–550.
- Gibbs, Daniel. 2025. “A Model of Policy Timing: Hedging, Pandering, and Gambling.” *American Journal of Political Science* . Online first.
URL: <https://doi.org/10.1111/ajps.12993>
- Gieczewski, Germán and Christopher Li. 2022. “Dynamic Policy Sabotage.” *American Journal of Political Science* 66(3):617–629.
- Gratton, Gabriele, Richard Holden and Anton Kolotilin. 2018. “When to Drop a Bombshell.” *The Review of Economic Studies* 85(4):2139–2172.

- Guttman, Ilan, Ilan Kremer and Andrzej Skrzypacz. 2014. "Not Only What but Also When: A Theory of Dynamic Voluntary Disclosure." *American Economic Review* 104(8):2400–2420.
- Hacker, Jacob S. 1999. *The Road to Nowhere: The Genesis of President Clinton's Plan for Health Security*. Princeton, NJ: Princeton University Press.
- Hitt, Matthew P., Craig Volden and Alan E. Wiseman. 2017. "Spatial Models of Legislative Effectiveness." *American Journal of Political Science* 61(3):575–590.
- Judd, Gleason. 2017. "Showing Off: Promise and Peril in Unilateral Policymaking." *Quarterly Journal of Political Science* 12(2):241–268.
- Kingdon, John W. 2003. *Agendas, Alternatives, and Public Policies*. 2 ed. New York: Longman.
- Krehbiel, Jay N. 2021. "Strategic Delay and the Use of Incompatibility Rulings at the German Constitutional Court." *The Journal of Politics* 83(3):821–833.
- Majumdar, Sumon and Sharun W Mukand. 2004. "Policy gambles." *American Economic Review* 94(4):1207–1222.
- Maskin, Eric and Jean Tirole. 2004. "The Politician and the Judge: Accountability in Government." *American Economic Review* 94(4):1034–1054.
- Ostrander, Ian. 2016. "The Logic of Collective Inaction: Senatorial Delay in Executive Nominations." *American Journal of Political Science* 60(4):1063–1076.
- Potter, Rachel Augustine. 2017. "Slow-Rolling, Fast-Tracking, and the Pace of Bureaucratic Decisions in Rulemaking." *The Journal of Politics* 79(3):841–855.
- Shaver, Benjamin. 2025. "Signaling Ability through Policy Change." *The Journal of Politics* . Forthcoming.

Thrower, Sharece. 2018. “Policy Disruption through Regulatory Delay in the Trump Administration.” *Presidential Studies Quarterly* 48(3):517–536.

Zhou, Beixi. 2024. “Optimal Disclosure Windows.” Working paper.

A Online Appendix

Overview

This Appendix contains proofs of all formal claims in the main text: Lemmas 1-2 and Propositions 1-8.¹⁴ We also state and prove three additional lemmas not stated in the main text, Lemmas 3-5, which we use in our proofs. The Appendix is organized as follows. We first prove Lemma 1. We then state and prove Lemma 3 which characterizes and proves the existence of an equilibrium in which the high type chooses $a_2 = 1$ with probability one in $t = 2$ when $q_2 = 0$. Next we state and prove Lemma 4 which fully characterizes the set of all equilibria, identifies the unique welfare maximizing equilibrium, and derives $b > \bar{b}$ as a necessary and sufficient condition for the set of equilibria to be a singleton containing only the equilibrium characterized in Lemma 3. Next we use Lemmas 3 and 4 to prove Lemma 2 and Proposition 1. At the end of our proof of Proposition 1, we derive a closed-form expression for $k(b)$ (Equation 8). We then state and prove Lemma 5 which characterizes how variation in λ affects $k(b)$ and is used throughout our subsequent proofs of comparative static results. The remainder of the Appendix contains proofs of Propositions 2-8 in their numbered order.

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¹⁴Remark 1 is an immediate corollary of Proposition 1. Each Corollary 1-4 follows directly from a proposition which we note in the main text where each corollary appears.

A.1 Proof of Lemma 1

We first show that there is no equilibrium in which $\gamma_1 = 0$. To do so, we consider $\gamma_1 = 0$ and show that the high type's first-period strategy must also play $a_1 = 1$ with probability zero, that is, $\Pr(a_1 = 1|\theta = H) = 0$. We then show that, given that neither type plays $a_1 = 1$, in order for equilibrium to survive D1, the low type must play $a_2 = 1$ with positive probability in equilibrium, that is, $\gamma_2 > 0$. It follows from this that a high type with a well-designed policy in $t = 2$ receives a strictly higher payoff from implementing the policy than a low type (since $\mu_\phi < 1$ by Bayes rule given $\gamma_2 > 0$ and because successful policy yields the high type with $q_2 = 1$ a reputation payoff of $1 > \mu_\phi$ which occurs with probability $\rho > 0$). We use this to pin down equilibrium continuation values in the first period for the low type and a high type with $q_1 = 1$. We then use these continuation values to show that the net gain from deviation by a high type with $q_1 = 1$ in $t = 1$ to $a_1 = 1$ is strictly greater than the net gain from deviation for a low type. Hence D1 requires $\mu_1 = 1 > \mu_2$. We complete the proof by observing that $\mu_1 \leq \mu_2$ is necessary to prevent deviation from the low type which implies that no equilibrium with $\gamma_1 = 0$ survives D1.

Proof that $\gamma_1 > 0$.

Suppose that $\gamma_1 = 0$ in equilibrium.

Step 1: Show $\Pr(a_1 = 1|\theta = H) = 0$.

The low type's equilibrium payoff from $a_1 = 1$ is $b + \mu_1$ and his equilibrium payoff from $a_1 = 0$ is $\max\{b + \mu_2, \mu_\phi\}$. Thus $\gamma_1 = 0$ in equilibrium requires that

$$b + \mu_1 \leq \max\{b + \mu_2, \mu_\phi\}.$$

If $\Pr(a_1 = 1|\theta = H) > 0$, then $\mu_1 = 1$ by Bayes' rule. Hence if $\Pr(a_1 = 1|\theta = H) > 0$, it must be that $b + 1 \leq \max\{b + \mu_2, \mu_\phi\}$. Since $b + 1 > 1$ and $\mu_\phi \leq 1$, it follows that $\max\{b + \mu_2, \mu_\phi\} = b + \mu_2 = b + 1$. Since $b + \mu_2 = b + 1 > \mu_\phi$, the low type's equilibrium strategy, given $\gamma_1 = 0$ and $\Pr(a_1 = 1|\theta = H) > 0$, must play $a_1 = 0, a_2 = 1$ with probability one, i.e., $\gamma_2 = 1$. But then $\mu_2 < 1$ by Bayes' rule which implies a profitable deviation for the low type from $a_1 = 0, a_2 = 1$ to $a_1 = 1$. Hence if $\gamma_1 = 0$ in equilibrium, then $\Pr(a_1 = 1|\theta = H) = 0$ too. Thus in any equilibrium with $\gamma_1 = 0$, it must be that $a_1 = 1$ is off the equilibrium path.

Step 2: Show $\gamma_2 > 0$.

In $t = 2$, for both types, $a_2 = 0$ yields a payoff of μ_ϕ . For a high type with $q_2 = 0$ and a low type, $a_2 = 1$ yields a payoff of $b + \mu_2$. For a high type with $q_2 = 1$, choosing $a_2 = 1$ provides an expected payoff of $b + \rho + (1 - \rho)\mu_2$ since $\Pr(\theta = H|s = 1) = 1$ in every equilibrium.

If $\gamma_2 = 0$ in equilibrium and the high type plays $a_2 = 1$ with positive probability, then $\mu_\phi < 1$ and $b + \mu_2 = b + 1$ by Bayes' rule. But then there is clearly a profitable deviation from $a_2 = 0$ to $a_2 = 1$ for the low type since $b + \mu_2 > \mu_\phi$. Hence if $\gamma_2 = 0$ in equilibrium, then $\Pr(a_2 = 1|\theta = H) = 0$ too. In this case, the only action profile on the equilibrium path is $a_1 = a_2 = 0$. By Bayes' rule $\mu_\phi = p$. The net gain for a high type with $q_2 = 1$ from deviating from $a_2 = 0$ to $a_2 = 1$ is $b + \rho + (1 - \rho)\mu_2 - p$ whereas the same deviation yields a net gain of $b + \mu_2 - p$ for a low type. Note that $\mu_2 < 1$ is necessary to prevent deviations from both types. Since $b + \rho + (1 - \rho)\mu_2 - p - [b + \mu_2 - p] = \rho(1 - \mu_2) > 0$ for all $\mu_2 < 1$, D1 requires the voter to infer from $a_2 = 1$ that the incumbent is a high type with $q_2 = 1$. Hence $\mu_2 = 1$ under D1 in any equilibrium with $\gamma_1 = \gamma_2 = 0$. But then the low type has a profitable deviation from $a_1 = a_2 = 0$ to $a_1 = 0, a_2 = 1$. Thus for any equilibrium in which $\gamma_1 = 0$, surviving D1 requires $\gamma_2 > 0$.

Step 3: Derive Equilibrium Continuation Values.

Notice that $\gamma_2 > 0$ implies that the low type's equilibrium payoff is $b + \mu_2 \in [\mu_\phi, b + 1)$ where $\mu_2 < 1$ by Bayes' rule and $b + \mu_2 \geq \mu_\phi$ follows from the sequential rationality of the low type's strategy. This further implies that $\Pr(a_2 = 1|\theta = H, q_2 = 1, a_1 = 0) = 1$ since $b + \rho + (1 - \rho)\mu_2 > b + \mu_2 \geq \mu_\phi$. Let $v_L = b + \mu_2$, $v_{H1} = b + \rho + (1 - \rho)\mu_2$, and $v_{H0} = \lambda v_{H1} + (1 - \lambda)v_L$ denote the equilibrium continuation value in $t = 1$ for a low type, high type with $q_1 = 1$, and high type with $q_1 = 0$, respectively. Note that $1 + b > v_{H1} > v_{H0} > v_L \geq \mu_\phi$.

Step 4: Equilibrium requires $\mu_1 \leq \mu_2 < 1$ but D1 requires $\mu_1 = 1$.

In $t = 1$, a low type and a high type with $q_1 = 0$ receive a payoff of $b + \mu_1$ if they deviate to $a_1 = 1$ whereas a high type with $q_1 = 1$ receives a deviation payoff of $b + (1 - (1 - \rho)^2) + (1 - \rho)^2\mu_2$. The net gains (relative to expected equilibrium payoff from choosing $a_1 = 0$) for a low type, a high type with $q_1 = 0$, and a high type with $q_1 = 1$ respectively from deviating

to $a_1 = 1$ are therefore

$$\begin{aligned} & b + \mu_1 - v_L, \\ & b + \mu_1 - v_{H0}, \text{ and} \\ & b + \mu_1(1 - \rho)^2 + 1 - (1 - \rho)^2 - v_{H1}. \end{aligned}$$

Since $v_L < v_{H0}$, it follows that $b + \mu_1 - v_L > b + \mu_1 - v_{H0}$. Thus under D1, the voter infers that a deviation from $a_1 = 0$ to $a_1 = 1$ is weakly more likely to come from a low type than a high type with $q_1 = 0$. Therefore the relevant comparison under D1 for a deviation from $a_1 = 0$ to $a_1 = 1$ is between a high type with $q_1 = 1$ and a low type. The difference in net gains from deviation between a high type with $q_1 = 1$ and a low type is

$$\begin{aligned} & [b + \mu_1(1 - \rho)^2 + 1 - (1 - \rho)^2 - v_{H1}] - [b + \mu_1 - v_L] \\ & = [1 - (1 - \rho)^2](1 - \mu_1) + v_L - v_{H1} = [1 - (1 - \rho)^2](1 - \mu_1) - (1 - \rho)\mu_2 \end{aligned}$$

If $[1 - (1 - \rho)^2](1 - \mu_1) - (1 - \rho)\mu_2 > 0$, then D1 requires $\mu_1 = 1$ since the inequality implies that any deviation to $a_1 = 1$ is more likely to come from a high type with $q_1 = 1$ than a low type. Given $\mu_1 = 1$, then each type receives a payoff of $1 + b$ from choosing $a_1 = 1$. Because $v_L < 1 + b$, a profitable deviation exists for the low type. Hence an equilibrium with $\gamma_1 = 0$ does not survive D1 if $[1 - (1 - \rho)^2](1 - \mu_1) - (1 - \rho)\mu_2 > 0$ for some $\mu_1 < 1$. We now show that this inequality must be satisfied.

To prevent deviation from the low type in $t = 1$, it must be that $\mu_1 \leq \mu_2$. Thus minimizing the positive term in the inequality subject to this equilibrium condition yields

$$[1 - (1 - \rho)^2](1 - \mu_2) - (1 - \rho)\mu_2 = 1 - (1 - \rho)^2 - \mu_2[1 - (1 - \rho)^2 - (1 - \rho)]$$

Since $1 - (1 - \rho)^2 > 1 - (1 - \rho)^2 - (1 - \rho)$ and $\mu_2 < 1$, it follows that

$$1 - (1 - \rho)^2 - \mu_2[1 - (1 - \rho)^2 - (1 - \rho)] > 0.$$

Thus there is no equilibrium with $\gamma_1 = 0$ that survives D1. ■

An analogous proof shows that there is no equilibrium in which $\gamma_2 = 0$. Since $\gamma_1 > 0$ and $\gamma_2 > 0$ in equilibrium, it follows that $\gamma_1 \in (0, 1)$ and $\gamma_2 \in (0, 1)$.

We now show that $\Pr(a_t = 1 | \theta = H, q_t = 1) = 1$ for $t \in \{1, 2\}$. Consider a high type in $t = 2$ with $q_2 = 1$. The low type's strategy, $\gamma_1 \in (0, 1)$ and $\gamma_2 \in (0, 1)$, implies that $1 + b > b + \mu_2 \geq \mu_\phi$ since $\mu_2 < 1$ by Bayes' rule and the low type must weakly prefer $a_2 = 1$

to $a_2 = 0$. Thus

$$\begin{aligned}
EU[a_2 = 1 | \theta = H, q_2 = 1, a_1 = 0] &= b + \rho + (1 - \rho)\mu_2 \\
&> b + \mu_2 \geq \mu_\phi \\
&= EU[a_2 = 0 | \theta = H, q_2 = 1, a_1 = 0].
\end{aligned}$$

Hence the high type must play $a_2 = 1$ in $t = 2$ with probability one in equilibrium if $q_2 = 1$. Now consider a high type in $t = 1$ with $q_1 = 1$. Since $q_1 = 1$ implies $q_2 = 1$ if $a_1 = 0$, the high type must choose $a_2 = 1$ in $t = 2$ if $a_1 = 1$. Moreover, $\gamma_1 \in (0, 1)$ and $\gamma_2 \in (0, 1)$ imply that $b + \mu_1 = b + \mu_2 < b + 1$. Thus,

$$\begin{aligned}
EU[a_1 = 1 | \theta = H, q_1 = 1] &= b + 1 - (1 - \rho^2) + (1 - \rho)^2\mu_1 \\
&> b + \rho + (1 - \rho)\mu_1 \\
&= b + \rho + (1 - \rho)\mu_2 \\
&= EU[a_2 = 1 | \theta = H, q_2 = 1, a_1 = 0] \\
&= EU[a_1 = 0 | \theta = H, q_1 = 1].
\end{aligned}$$

Therefore the high type must play $a_1 = 1$ with probability one in equilibrium if $q_1 = 1$.

Finally, we show that $\Pr(a_1 = 1 | \theta = H, q_1 = 0) = 0$. To do so, we need to characterize the high type's continuation value from choosing $a_1 = 0$ when $q_1 = 0$. With probability λ , he receives $q_2 = 1$ in $t = 2$ in which case he plays $a_2 = 1$ with probability one which yields a payoff of $b + \rho + (1 - \rho)^2\mu_2$. With probability $1 - \lambda$, the high type receives $q_2 = 0$. For a high type in $t = 2$ with $q_2 = 0$, his expected payoff from each action is identical to the low type's. Since $\gamma_1 \in (0, 1)$ and $\gamma_2 \in (0, 1)$, the low type's expected payoff in $t = 2$ must be $b + \mu_2 = b + \mu_1$. Thus the high type's expected payoff in $t = 2$ when $q_2 = 0$ is also $b + \mu_2 = b + \mu_1$. Thus:

$$\begin{aligned}
EU[a_1 = 1 | \theta = H, q_1 = 0] &= b + \mu_1 \\
&< \lambda[b + \rho + (1 - \rho)\mu_1] + (1 - \lambda)[b + \mu_1] \\
&= \lambda[b + \rho + (1 - \rho)\mu_2] + (1 - \lambda)[b + \mu_2] \\
&= \lambda EU[a_2 = 1 | \theta = H, q_2 = 1, a_1 = 0] \\
&\quad + (1 - \lambda) EU[a_2 = 1 | \theta = H, q_2 = 0, a_1 = 0] \\
&= EU[a_1 = 0 | \theta = H, q_1 = 0].
\end{aligned}$$

Thus $\Pr(a_1 = 1 | \theta = H, q_1 = 0) = 0$ in equilibrium. $\square$

A.2 Lemmas 3 and 4

Lemma 3. *For all b , there exists an equilibrium in which the high type chooses $a_2 = 1$ with probability one if $q_2 = 0$, $\gamma_1 = \frac{\lambda(\rho-1)^2}{\lambda^2\rho + \lambda(\rho-3)\rho+1}$, and $\gamma_2 = 1 - \gamma_1$.*

Proof: It follows from Lemma 1 that if the high type chooses $a_2 = 1$ with probability one when $q_2 = 0$ in equilibrium, then the low type's strategy must satisfy

$$b + \mu_1 = b + \mu_2 \geq \mu_\phi.$$

If $\gamma_1 + \gamma_2 < 1$, then $\mu_\phi = 0$ by Bayes' rule. Since the high type's strategy plays both $a_1 = 1$ and $(a_1, a_2) = (0, 1)$ with positive probability, Bayes' rule implies $\mu_1 = \mu_2 > 0$. The low type must therefore play $(a_1, a_2) = 0, 0$ with probability zero. Otherwise, a profitable deviation from $(a_1, a_2) = (0, 0)$ to either $a_1 = 1$ or $(a_1, a_2) = (0, 1)$ exists for the low type. Thus in equilibrium, $\gamma_1 = 1 - \gamma_2$. The low type's indifference condition, $\mu_1 = \mu_2$, is therefore equivalent to

$$\frac{p\lambda(1-\rho)^2}{p\lambda(1-\rho)^2 + (1-p)\gamma_1} = \frac{p(1-\lambda)[\lambda(1-\rho) + (1-\lambda)]}{p(1-\lambda)[\lambda(1-\rho) + (1-\lambda)] + (1-p)(1-\gamma_1)}.$$

Solving explicitly for γ_1 yields the expression in Lemma 3. It is straightforward to check that $\gamma_1 \in (0, 1)$ and that γ_1 is independent of b . Thus the equilibrium exists for all b . $\square$

Lemma 4. *A unique welfare-maximizing equilibrium exists. If $b \leq \bar{b}$, then in the unique welfare-maximizing equilibrium, the high type chooses $a_2 = 0$ with probability one if $q_2 = 0$, $\gamma_1 \in (0, 1)$, and $\gamma_2 \in (0, 1)$; moreover, $1 - \gamma_1 - \gamma_2 > 0$ if $b < \bar{b}$. If $b > \bar{b}$, then the unique welfare-maximizing equilibrium is characterized by Lemma 3.*

Proof

It follows from Lemma 1 that strategy profiles in equilibrium can vary only with the low type's strategy, characterized by (γ_1, γ_2) , and the high type's strategy in $t = 2$ when $q_2 = 0$,

$$\omega \equiv Pr(a_2 = 1 | \theta = H, q_2 = 0, a_1 = 0).$$

Lemma 1 also implies that the voter's equilibrium welfare is

$$V^* = p(1 - (1 - \lambda)^2) - \zeta p(1 - \lambda)^2 \omega - \zeta(1 - p)(\gamma_1 + \gamma_2)$$

Additionally, since both the low and high type choose $a_1 = 1$ and $(a_1, a_2) = (0, 1)$ with

positive probability in equilibrium, Lemma 1 implies

$$\begin{aligned}\mu_1 &= \left[1 + \left(\frac{1-p}{p} \right) \left(\frac{\gamma_1}{\lambda(1-\rho)^2} \right) \right]^{-1}, \\ \mu_2 &= \left[1 + \left(\frac{1-p}{p} \right) \left(\frac{\gamma_2}{(1-\lambda)[\lambda(1-\rho) + \omega(1-\lambda)]} \right) \right]^{-1}.\end{aligned}$$

Because the low type's strategy requires that $b + \mu_1 = b + \mu_2$, we can express γ_2 in terms of ω and γ_1 as

$$\gamma_2(\gamma_1, \omega) = \gamma_1 \left(\frac{1-\lambda}{1-\rho} \right) \left[1 + \frac{\omega}{\lambda} \left(\frac{1-\lambda}{1-\rho} \right) \right]$$

which is obtained by solving $\mu_1 = \mu_2$ for γ_2 . Substituting $\gamma_2(\gamma_1, \omega)$ into μ_1 and μ_2 yields

$$\mu_1 = \mu_2 = \left[1 + \left(\frac{1-p}{p} \right) \left(\frac{\gamma_1}{\lambda(1-\rho)^2} \right) \right]^{-1}$$

Additionally, if $1 - \gamma_1 - \gamma_2 + \omega > 0$, then μ_ϕ is determined by Bayes' rule. Substituting $\gamma_2(\gamma_1, \omega)$ into μ_ϕ yields

$$\mu_\phi = \left[1 + \left(\frac{1-p}{p} \right) \left(\frac{1}{(1-\lambda)^2(1-\omega)} \right) \left(1 - \gamma_1 \left(1 + \left(\frac{1-\lambda}{1-\rho} \right) + \frac{\omega}{\lambda} \left(\frac{1-\lambda}{1-\rho} \right)^2 \right) \right) \right]^{-1}$$

if $\omega < 1$ and $\mu_\phi = 0$ otherwise.

Substituting the above expressions for $\mu_1 = \mu_2$ and μ_ϕ into the indifference condition $b + \mu_1 = \mu_\phi$ and rearranging yields a quadratic equation in γ_1 ,

$$f(\gamma_1) = \gamma_1^2 P^2 Z(\omega) + \gamma_1 [P(1-b)Y(\omega) + \lambda(1-\rho)^2(1+b)Z(\omega) - bP^2] - \lambda(1-\rho)^2 [bY(\omega) + P(1+b)] = 0$$

where

$$\begin{aligned}Y(\omega) &= (1-\lambda)^2(1-\omega), \\ Z(\omega) &= \left(1 + \left(\frac{1-\lambda}{1-\rho} \right) + \frac{\omega}{\lambda} \left(\frac{1-\lambda}{1-\rho} \right)^2 \right), \text{ and} \\ P &= \frac{1-p}{p}\end{aligned}$$

The positive leading coefficient, $P^2 Z(\omega) > 0$, in $f(\gamma_1)$ implies that $f(\gamma_1)$ is convex and $f(0) < 0$ implies that there exist two real solutions to $f(\gamma_1) = 0$, one negative and one

positive. It follows that $\gamma_1 \in (0, 1)$ is uniquely characterized by

$$\gamma_1 = \gamma_1(\omega, b) \equiv \left(\frac{-\beta(\omega, b) + \sqrt{D(\omega, b)}}{2Z(\omega)P^2} \right)$$

where

$$\begin{aligned} \beta(\omega, b) &= P(1 - b)Y(\omega) + \lambda(1 - \rho)^2(1 + b)Z(\omega) - bP^2, \text{ and} \\ D(\omega, b) &= [\beta(\omega, b)]^2 + 4P^2\lambda(1 - \rho)^2[bY(\omega) + P(1 + b)]Z(\omega). \end{aligned}$$

Note that for a given ω ,

$$\gamma_1 + \gamma_2 = \gamma_1(\omega, b) + \gamma_2(\gamma_1(\omega, b), \omega) = \gamma_1(\omega, b)Z(\omega)$$

Direct computations show that $\gamma_1(0, b)Z(0) > 0$ and that $\gamma_1(\omega, b)Z(\omega)$ is strictly increasing in ω . Solving $\gamma_1(\omega, b)Z(\omega) = 1$ for ω yields a unique solution,

$$\eta(b) = \frac{b(\lambda^2(-p)(\rho - 1) - \lambda p(\rho^2 - 3\rho + 2) + p - 1) - p + 1}{b(\lambda - 1)^2 p}.$$

It is straightforward to check that $\eta(b)$ is strictly decreasing in b . Thus if $\eta(b) > 0$, there exists an equilibrium for each $\omega \in [0, \eta(b)]$. Solving $\eta(b) = 1$ for b yields

$$\underline{b} \equiv \frac{1 - p}{\lambda p \rho (\lambda + \rho - 3) + 1}.$$

Thus if $b \leq \underline{b}$, then all $\omega \in [0, 1]$ are consistent with equilibrium. Moreover, for all $b \leq \underline{b}$, the equilibrium for $\omega = 1$ corresponds to the equilibrium characterized in Lemma 3. Solving $\eta(b) = 0$ for b yields

$$\bar{b} = \frac{1 - p}{(1 - p) + p\lambda(1 - \rho)(2 - \lambda - \rho)} > \underline{b}.$$

Thus for $b \in (\underline{b}, \bar{b}]$, $\omega < 1$ characterizes an equilibrium if and only if $\omega \in [0, \eta(b)]$. If $b > \bar{b}$, then $\gamma_1(\omega, b)Z(\omega) > 1$ for all $\omega \in [0, 1]$ which implies that no $\omega < 1$ is consistent equilibrium. Thus the existence of any equilibrium with $\omega < 1$ implies the existence of an equilibrium with $\omega = 0$ and such an equilibrium exists if and only if $b \leq \bar{b}$.

Because $\gamma_1 + \gamma_2 = \gamma_1(\omega, b)Z(\omega)$ is increasing in ω , it follows that $\omega = 0$ characterizes the unique welfare-maximizing equilibrium if $b \leq \bar{b}$. If $b > \bar{b}$, then the set of equilibria is a singleton, containing only the equilibrium characterized in Lemma 3 and is therefore, trivially, the welfare-maximizing equilibrium. $\square$

A.3 Proof of Lemma 2

Under Assumption 1, the result follows directly from Lemma 4. $\square$

A.4 Proof of Proposition 1

Under Assumption 1, it follows from Lemma 4 that in the unique welfare maximizing equilibrium,

$$\mu_1 + b = \mu_2 + b = \mu_\phi$$

where

$$\mu_1 = \frac{ph_1}{ph_1 + (1-p)\gamma_1}, \quad \mu_2 = \frac{ph_2}{ph_2 + (1-p)\gamma_2}, \quad \mu_\phi = \frac{ph_\phi}{ph_\phi + (1-p)\gamma_\phi}.$$

Note that

$$\mu_1 + b = \mu_2 + b \implies \frac{\gamma_1}{\gamma_2} = \frac{h_1}{h_2}.$$

Since $\gamma_\phi = 1 - \gamma_1 - \gamma_2 \in (0, 1)$, it follows that

$$\gamma_1 = kh_1, \quad \gamma_2 = kh_2, \quad \gamma_\phi = 1 - k\mathcal{H}_{12}$$

for some $k \in [0, \mathcal{H}^{-1})$ where $\mathcal{H}_{12} = (h_1 + h_2)$. Substituting $\gamma_1 = kh_1$ into μ_1 and $\gamma_2 = kh_2$ into μ_2 yields

$$\mu_1 = \mu_2 = \mu_R = \frac{p}{p + (1-p)k}.$$

Thus $b + \mu_R = \mu_\phi$ becomes

$$b + \frac{p}{p + (1-p)k} = \frac{ph_\phi}{ph_\phi + (1-p)\gamma_\phi}.$$

Rearranging and simplifying yields a quadratic equation in k ,

$$G \equiv c_2 k^2 + c_1 k + c_0 = 0 \tag{4}$$

where

$$c_0 = p(b + (1-p) - bp(1 - h_\phi)) \tag{5}$$

$$c_1 = -(1-p)(bp(1 + \mathcal{H}_{12} - h_\phi) + p\mathcal{H} - b) \tag{6}$$

$$c_2 = -\mathcal{H}_{12}b(1-p)^2. \tag{7}$$

Analyzing the roots of the equation shows that for every $b \in [0, \bar{b}]$,

$$k = k(b) \equiv \frac{-c_1 - \sqrt{\Delta}}{2c_2}, \quad \text{where } \Delta \equiv c_1^2 - 4c_2c_0, \quad (8)$$

is the unique root such that $k \in [0, \mathcal{H}_{12}^{-1}]$ and that $k(b) : [0, \bar{b}] \rightarrow [\mathcal{H}^{-1}, \mathcal{H}_{12}^{-1}]$ is a strictly increasing bijection. $\square$

A.5 Lemma 5

Lemma 5. *If b is sufficiently low, then $k(b)$ is monotonically increasing in λ . If b is sufficiently high, then $k(b)$ is decreasing in λ if λ is sufficiently low and increasing in λ if λ is sufficiently high.*

Proof:

From (4), $k(b)$ is implicitly characterized by $G = 0$. By the implicit function theorem,

$$k_\lambda = -\frac{G_\lambda}{G_k}. \quad (9)$$

Since $k(b) = (-c_1 - \sqrt{\Delta})/(2c_2)$ by (8), the derivative of G with respect to k evaluated at $k = k(b)$ satisfies the quadratic identity

$$G_k = 2c_2k + c_1 = -\sqrt{\Delta} < 0. \quad (10)$$

Thus (9)-(10) imply that

$$\text{sign}(k_\lambda(b)) = \text{sign}(G_\lambda|_{k=k(b)}). \quad (11)$$

Therefore to establish the sign of k_λ , it is sufficient to identify the sign of G_λ evaluated at $k = k(b)$.

Because c_2 , c_1 , and c_0 are continuous in (b, λ) , and because $\Delta > 0$ for $b < \bar{b}$, it follows that $k(b)$ is continuous in (b, λ) . Since G_λ is continuous (λ, k, p) , the mapping

$$(b, \lambda) \mapsto G_\lambda|_{k=k(b)}$$

is too on $(0, \bar{b}) \times (0, 1)$.

Next, direct calculations show that for all $b < \bar{b}$,

$$\lim_{\lambda \rightarrow 1} G_\lambda|_{k=k(b)} = \rho(1 - \rho)(1 - p)[b(1 - p)k(b)^2 + p(1 + b)k(b)] > 0$$

since $k(b) > 0$. By continuity, there exists $\lambda_H < 1$ such that $G_\lambda|_{k=k(b)} > 0$ for all $\lambda \in (\lambda_H, 1)$.

Now consider $b > p$. As $\lambda \downarrow 0$, $h_\phi = (1 - \lambda)^2 \rightarrow 1$ and $\mathcal{H}_{12} \sim (1 - \rho)(2 - \rho)\lambda$. Since $b > p$, we have $c_1 = (1 - p)(b - p) > 0$ while $c_2 \rightarrow 0^-$. Thus $k(b)$ satisfies

$$k(b) \sim \frac{b - p}{b(1 - \rho)(2 - \rho)} \cdot \frac{1}{\lambda} \quad (\lambda \downarrow 0, b > p).$$

Substituting into $G_\lambda|_{k=k(b)}$ yields the leading order expansion

$$G_\lambda|_{k=k(b)} = -\frac{(b - p)^2}{b(1 - \rho)(2 - \rho)} \frac{1}{\lambda^2} + O\left(\frac{1}{\lambda}\right) < 0$$

for all sufficiently small λ whenever $b > p$. Hence $k(b)$ is strictly decreasing for all λ in a neighborhood $(0, \lambda_L)$ if $b > p$.¹⁵ $\square$

A.6 Proof of Proposition 2

We first prove the result for λ . Since $h_\phi = (1 - \lambda)^2$ is strictly decreasing in λ , it follows immediately that $1 - h_\phi$ is strictly increasing. Moreover, $h_1 + h_2 = \lambda(1 - \rho)^2 + \lambda(1 - \lambda)(1 - \rho)$ is strictly increasing. In equilibrium, the low type's strategy must satisfy $b + \mu_R = \mu_\phi$, which can be expressed as

$$b + \frac{p(h_1 + h_2)}{p(h_1 + h_2) + (1 - p)(1 - \gamma_\phi)} = \frac{ph_\phi}{ph_\phi + (1 - p)\gamma_\phi}.$$

Note that h_ϕ is strictly decreasing and $h_1 + h_2$ strictly increasing in λ . This implies that, holding γ_ϕ fixed, as λ increases, μ_ϕ decreases and μ_R increases. Since μ_ϕ is strictly decreasing and μ_R strictly increasing in γ_ϕ , it follows that γ_ϕ must strictly decrease in order to maintain equality. Since $1 - \gamma_\phi = 1 - (\gamma_1 + \gamma_2)$, this implies that $\gamma_1 + \gamma_2$ is strictly increasing in λ .

Next we prove the result for ρ . As ρ increases, $h_1 + h_2$ strictly decrease while h_ϕ remains constant. Thus, holding γ_ϕ constant, μ_R strictly decreases and μ_ϕ remains constant if ρ increases. Since μ_R strictly increases and μ_ϕ strictly decreases in γ_ϕ , it must be that γ_ϕ increases in order to reestablish equality. Therefore $\gamma_1 + \gamma_2 = 1 - \gamma_\phi$ is strictly decreasing in ρ .

Finally, we prove the result for b . Proposition 1 establishes that in equilibrium, $\gamma_1 + \gamma_2 = k(b)(h_1 + h_2)$ and that $k(b)$ is strictly increasing in b . Since b enters $\gamma_1 + \gamma_2$ only through $k(b)$ and $h_1 + h_2 > 0$, it follows that $\gamma_1 + \gamma_2$ is strictly increasing in b . $\square$

¹⁵This condition is sufficient, not necessary; if $\bar{b}|_{\lambda=0} < p$, the nonmonotonicity can still obtain, but we do not pursue the sharper b threshold here.

A.7 Proof of Proposition 3

Given the high type's equilibrium strategy, the odds that the high type implements policy in $t = 1$, conditional on him implementing the policy in either period, are

$$\begin{aligned}\Omega_H &\equiv \frac{\Pr(a_1 = 1 | \theta = H, (a_1, a_2) \neq (0, 0))}{\Pr((a_1, a_2) = (0, 1) | \theta = H, (a_1, a_2) \neq (0, 0))} \\ &= \left(\frac{\lambda}{\lambda + (1 - \lambda)\lambda} \right) \left(\frac{\lambda(1 - \lambda)}{\lambda + (1 - \lambda)\lambda} \right)^{-1} \\ &= \frac{1}{1 - \lambda}.\end{aligned}$$

Clearly Ω_H is constant in ρ and b and strictly increasing in λ . Analogously, for the low type,

$$\Omega_L \equiv \frac{\Pr(a_1 = 1 | \theta = L, (a_1, a_2) \neq (0, 0))}{\Pr((a_1, a_2) = (0, 1) | \theta = L, (a_1, a_2) \neq (0, 0))} = \frac{k(b)h_1}{k(b)h_2} = \frac{1 - \rho}{1 - \lambda}.$$

Thus Ω_L is increasing in λ , decreasing in ρ , and constant in b .

The odds that the politician, unconditional on type, implements policy in $t = 1$, conditional on him implementing the policy in either period, are

$$\begin{aligned}\Omega &= \frac{\Pr(a_1 = 1 | (a_1, a_2) \neq (0, 0))}{\Pr((a_1, a_2) = (0, 1) | (a_1, a_2) \neq (0, 0))} = \frac{p\lambda + (1 - p)k(b)h_1}{p(1 - \lambda)\lambda + (1 - p)k(b)h_2} \\ &= \left(\frac{p + (1 - p)k(b)(1 - \rho)^2}{p + (1 - p)k(b)(1 - \rho)} \right) \left(\frac{1}{1 - \lambda} \right)\end{aligned}$$

Then

$$\frac{\partial \Omega}{\partial b} = - \left(\frac{p(1 - p)(1 - \rho)\rho}{(1 - \lambda)[p + (1 - p)k(b)(1 - \rho)]^2} \right) \left(\frac{\partial k(b)}{\partial b} \right) < 0$$

since $k(b)$ is increasing in b by Proposition 1. $\square$

A.8 Proof of Proposition 4

Define $\Pr(\text{well-designed} | a_t = 1)$ as the probability that a policy implemented in period t is well-designed. Using equilibrium strategies, these probabilities are:

$$\Pr(\text{well-designed} | a_1 = 1) = \frac{p\lambda}{p\lambda + (1 - p)\gamma_1} = \frac{p}{p + \left(\frac{1 - p}{k(b)\mathcal{H}} \right) (1 - \rho)^2},$$

$$\Pr(\text{well-designed} | a_1 = 0, a_2 = 1) = \frac{p(1 - \lambda)\lambda}{p(1 - \lambda)\lambda + (1 - p)\gamma_2} = \frac{p}{p + \left(\frac{1 - p}{k(b)\mathcal{H}} \right) (1 - \rho)}.$$

We then have

$$\Omega_1 = \frac{\Pr(\text{well-designed} \mid a_1 = 1)}{1 - \Pr(\text{well-designed} \mid a_1 = 1)} = \frac{p\lambda}{(1-p)\gamma_1}$$

and

$$\Omega_2 = \frac{\Pr(\text{well-designed} \mid a_1 = 0, a_2 = 1)}{1 - \Pr(\text{well-designed} \mid a_1 = 0, a_2 = 1)} = \frac{p(1-\lambda)\lambda}{(1-p)\gamma_2}.$$

Therefore

$$\frac{\Omega_1}{\Omega_2} = \frac{1}{1-\rho}.$$

Thus $\frac{\Omega_1}{\Omega_2} > 1$ which implies the first part of the proposition. Clearly $\frac{\Omega_1}{\Omega_2}$ is constant in b and λ and strictly increasing in ρ , thus establishing the second part. $\square$

A.9 Proof of Proposition 5

We start by providing complete expressions for interim beliefs.

$$\hat{\mu}_1 = (1 + Pk(b)(1-\rho)^2)^{-1}$$

$$\hat{\mu}_\phi = \left(1 + P \left(\frac{1 - k(b)\lambda(1-\rho)^2}{1-\lambda} \right)\right)^{-1}$$

$$\hat{\mu}_2 = (1 + Pk(b)(1-\rho))^{-1}$$

For reference,

$$\mu_\phi = \left(1 + P \left(\frac{1 - k(b)\lambda(1-\rho)(2-\lambda-\rho)}{(1-\lambda)^2} \right)\right)^{-1}$$

$$\mu_R = (1 + Pk(b))^{-1}$$

We prove the results for b sufficiently low by showing that the results hold for $b = 0$. Then the continuity of beliefs in b implies the result for $b \in (0, b_\varepsilon)$ for some $b_\varepsilon > 0$.

For $b = 0$, we have $k(0) = \mathcal{H}^{-1}$. Thus

$$\hat{\mu}_1 = \left(1 + P \left(\frac{(1-\rho)^2}{\mathcal{H}} \right)\right)^{-1}$$

$$\tilde{\mu}_1 = \left(1 + P \left(\frac{(1-\rho)}{\mathcal{H}} \right)\right)^{-1}$$

$$\hat{\mu}_2 = \left(1 + P \left(\frac{\lambda(1-\rho) + (1-\lambda)}{\mathcal{H}} \right)\right)^{-1}$$

$$\mu_R = \mu_\phi = \left(1 + P\left(\frac{1}{\mathcal{H}}\right)\right)^{-1}$$

Thus

$$(1 - \rho)^2 < (1 - \rho) < \lambda(1 - \rho) + (1 - \lambda) < 1$$

for all $\lambda \in (0, 1)$ and $\rho \in (0, 1)$ implies that

$$\hat{\mu}_1 > \hat{\mu}_2 > \hat{\mu}_\phi > \mu_R = \mu_\phi.$$

Next we show $\hat{\mu}_1 > p$ for all λ :

$$\hat{\mu}_1 > p \iff \frac{(1 - \rho)^2}{\mathcal{H}} < 1$$

and is straightforward to verify that

$$(1 - \rho)^2 - \mathcal{H} < 0$$

reduces to $\rho < 1$ which is true for all λ .

We prove that $\hat{\mu}_\phi < p$ for all λ similarly:

$$\hat{\mu}_\phi < p \iff \frac{\lambda(1 - \rho) + (1 - \lambda)}{\mathcal{H}} > 1$$

so

$$\lambda(1 - \rho) + (1 - \lambda) - \mathcal{H} = \lambda\rho(2 - \lambda - \rho) > 0$$

for all λ implies the result. It immediately follows that $\mu_\phi = \mu_R < p$ since $\hat{\mu}_\phi > \mu_\phi = \mu_R$.

Next we show that $\hat{\mu}_2 > p$ for sufficiently low λ and that $\hat{\mu}_2 < p$ for sufficiently high λ . We have that

$$\frac{(1 - \rho)}{\mathcal{H}} < 1 \implies (1 - \rho) - \mathcal{H} < 0.$$

Substituting and rearranging shows that

$$(1 - \rho) - \mathcal{H} = -\rho[\lambda^2 - \lambda(3 - \rho) + 1]$$

It is straightforward to check that $\lambda^2 - \lambda(3 - \rho) + 1 = 0$ has two real roots with a unique root in $(0, 1)$:

$$\lambda_- = \frac{3 - \rho - \sqrt{(3 - \rho)^2 - 4}}{2} \in (0, 1) \quad \text{for all } \rho \in (0, 1)$$

and

$$\lambda_+ = \frac{3 - \rho + \sqrt{(3 - \rho)^2 - 4}}{2} > 0 \quad \text{for all } \rho \in (0, 1).$$

Since $\lambda^2 - \lambda(3 - \rho) + 1$ has a positive leading coefficient, it follows that $\lambda^2 - \lambda(3 - \rho) + 1 > 0$ if $\lambda \in (0, \lambda_-)$ and $\lambda^2 - \lambda(3 - \rho) + 1 < 0$ if $\lambda \in (\lambda_-, 1)$. Thus $(1 - \rho) - \mathcal{H} < 0$ if $\lambda < \lambda_-$ and $(1 - \rho) - \mathcal{H} > 0$ otherwise which implies the result.

We now turn to the b sufficiently large case. We prove the result by showing that the inequalities hold for $b \uparrow \bar{b}$. The continuity of beliefs in $\bar{b}$ and λ then implies the result on $b \in (b', \bar{b})$ for some $b' \in (0, \bar{b})$.

Since $\lim_{b \rightarrow \bar{b}} k(b) = \mathcal{H}_{12}^{-1} = \frac{1}{(1 - \rho)\lambda(2 - \lambda - \rho)}$, it follows that $\lim_{b \rightarrow \bar{b}} \gamma_\phi = 0$. Hence $\mu_\phi \rightarrow 1$ as $b \rightarrow \bar{b}$.

Next we have

$$\begin{aligned} \lim_{b \rightarrow \bar{b}} \hat{\mu}_1 &= \left(1 + P \left(\frac{1 - \rho}{\lambda(2 - \lambda - \rho)} \right) \right)^{-1} \\ \lim_{b \rightarrow \bar{b}} \hat{\mu}_2 &= \left(1 + P \left(\frac{1}{\lambda(2 - \lambda - \rho)} \right) \right)^{-1} \\ \lim_{b \rightarrow \bar{b}} \hat{\mu}_\phi &= \left(1 + P \left(\frac{1}{2 - \lambda - \rho} \right) \right)^{-1} \\ \lim_{b \rightarrow \bar{b}} \mu_R &= \left(1 + P \left(\frac{1}{(1 - \rho)\lambda(2 - \lambda - \rho)} \right) \right)^{-1} \end{aligned}$$

We then have that

$$1, \frac{1 - \rho}{\lambda} < \frac{1}{\lambda} < \frac{1}{\lambda(1 - \rho)} \implies \hat{\mu}_1, \hat{\mu}_\phi > \hat{\mu}_2 > \mu_R$$

for all $\lambda \in (0, 1)$. Then

$$\frac{1}{\lambda(2 - \lambda - \rho)} > 1 \quad \text{for all } \lambda, \rho \in (0, 1)^2 \implies p > \hat{\mu}_2.$$

Next, $\hat{\mu}_1 < p$ if and only if $(1 - \rho) > \lambda(2 - \lambda - \rho) \implies \lambda < (1 - \rho)$. Similarly, $\hat{\mu}_\phi < p$ if and only if $1 > 2 - \lambda - \rho \implies \lambda > (1 - \rho)$. Thus $\hat{\mu}_1 < p < \hat{\mu}_\phi$ if $\lambda < (1 - \rho)$ and $\hat{\mu}_1 > p > \hat{\mu}_\phi$ if $\lambda > (1 - \rho)$. $\square$

A.10 Proof of Proposition 6

In equilibrium,

$$F \equiv b + \mu_R - \mu_\phi = 0$$

where

$$\mu_R = \frac{p(h_1 + h_2)}{p(h_1 + h_2) + (1-p)(\gamma_1 + \gamma_2)} = \frac{p(h_1 + h_2)}{p(h_1 + h_2) + (1-p)(h_1 + h_2)k} = \frac{p}{p + (1-p)k}$$

and

$$\mu_\phi = \frac{ph_\phi}{ph_\phi + (1-p)\gamma_\phi} = \frac{ph_\phi}{ph_\phi + (1-p)(1 - (h_1 + h_2)k)}.$$

To prove that μ_R and μ_ϕ are both decreasing in ρ , we first show that $k(b)$ is increasing in ρ . Note that $\frac{\partial \mu_R}{\partial k} < 0$ and $\frac{\partial \mu_\phi}{\partial k} > 0$. Therefore

$$F_k \equiv \frac{\partial F}{\partial k} = \frac{\partial \mu_R}{\partial k} - \frac{\partial \mu_\phi}{\partial k} < 0$$

Additionally, holding k fixed, $\frac{\partial \mu_\phi}{\partial \rho} < 0$ since $h_1 + h_2 = \lambda(1-\rho)(2-\lambda-\rho)$ and $h_\phi = (1-\lambda)^2$. Now applying the implicit function theorem to $F = 0$ shows that

$$\frac{\partial k}{\partial \rho} = -\left(\frac{1}{F_k}\right) \left(-\frac{\partial \mu_\phi}{\partial \rho}\right) = \left(\frac{1}{F_k}\right) \left(\frac{\partial \mu_\phi}{\partial \rho}\right) > 0$$

since $\frac{\partial \mu_\phi}{\partial \rho}|_k < 0$ and $F_k < 0$. Now totally differentiating μ_R respect to ρ , we have that

$$\frac{d\mu_R}{d\rho} = \frac{\partial \mu_R}{\partial \rho} + \frac{\partial \mu_R}{\partial k} \frac{\partial k}{\partial \rho} = \frac{\partial \mu_R}{\partial k} \frac{\partial k}{\partial \rho} < 0$$

since $\frac{\partial k}{\partial \rho} > 0$ and $\frac{\partial \mu_R}{\partial k} < 0$. Then, because $b + \mu_R = \mu_\phi$, it follows that

$$\frac{d\mu_\phi}{d\rho} = \frac{d\mu_R}{d\rho} < 0.$$

Next we show that μ_R is decreasing and μ_ϕ increasing in b . We establish in Proposition 1 that $k(b)$ is strictly increasing in b . Thus

$$\frac{d\mu_R}{db} = \frac{\partial \mu_R}{\partial k} \frac{\partial k}{\partial b} < 0$$

and

$$\frac{d\mu_\phi}{db} = \frac{\partial \mu_\phi}{\partial k} \frac{\partial k}{\partial b} > 0$$

because $\frac{\partial \mu_R}{\partial k} < 0$ and $\frac{\partial \mu_\phi}{\partial k} > 0$.

Finally, we prove comparative static results with respect to λ . Differentiating μ_R with

respect to λ yields

$$\frac{d\mu_R}{d\lambda} = \frac{\partial\mu_R}{\partial k} \frac{\partial k}{\partial\lambda}.$$

Then $b + \mu_R = \mu_\phi$ implies that

$$\frac{d\mu_\phi}{d\lambda} = \frac{d\mu_R}{d\lambda}.$$

Since $\frac{\partial\mu_R}{\partial k} < 0$, it follows that

$$\text{sign}\left(\frac{d\mu_R}{d\lambda}\right) = \text{sign}\left(\frac{d\mu_\phi}{d\lambda}\right) = -\text{sign}\left(\frac{\partial k}{\partial\lambda}\right).$$

The result then follows from Lemma 5. $\square$

A.11 Proof of Proposition 7

The voter's welfare is

$$V^* = p\lambda - \zeta(1-p)(\gamma_1 + \gamma_2)$$

so

$$\text{sign}\left(\frac{\partial V^*}{\partial\rho}\right) = -\text{sign}\left(\frac{\partial(\gamma_1 + \gamma_2)}{\partial\rho}\right)$$

and

$$\text{sign}\left(\frac{\partial V^*}{\partial b}\right) = -\text{sign}\left(\frac{\partial(\gamma_1 + \gamma_2)}{\partial b}\right)$$

By Proposition 2, $(\gamma_1 + \gamma_2)$ is strictly increasing in b and strictly decreasing in ρ . $\square$

A.12 Proof of Proposition 8

The result for the low type follows immediately from proposition 6. For the high type,

$$V_H^* = b(1 - h_\phi) + (1 - \mathcal{H}) + (h_1 + h_2)\mu_R + h_\phi\mu_\phi$$

We then have

$$\frac{\partial V_H^*}{\partial b} = (h_1 + h_2)\frac{\partial\mu_R}{\partial b} + h_\phi\frac{\partial\mu_\phi}{\partial b} + (1 - h_\phi) = 1 + \mathcal{H}\frac{\partial\mu_R}{\partial b}$$

where the second equality follows from $b + \mu_R = \mu_\phi \implies \mu'_\phi = 1 + \mu'_R$. Thus a sufficient condition for V_H^* to be increasing in b is $\frac{\partial\mu_R}{\partial b} > -1$. Suppose for contradiction that $\frac{\partial\mu_R}{\partial b} \leq -1$.

Since $b + \mu_R = \mu_\phi$ in equilibrium, this implies that

$$\frac{\partial \mu_\phi}{\partial b} = 1 + \frac{\partial \mu_R}{\partial b} \leq 0.$$

But we know from Proposition 6 that μ_ϕ is strictly increasing in b . Thus our assumption that $\frac{\partial \mu_R}{\partial b} \leq -1$ must be false. Therefore V_H^* is strictly increasing in b .

Next we show that V_H^* is strictly increasing in λ and ρ for $b = 0$. The smoothness of V_H^* in $\lambda, \rho \in (0, 1)^2$ then implies the result for sufficiently low b .

When $b = 0$, V_H^* is equivalent to

$$V_H^* = 1 - \mathcal{H} + \frac{p\mathcal{H}^2}{p\mathcal{H} + (1 - p)}.$$

We then have

$$\frac{\partial V_H^*}{\partial \mathcal{H}} = -\frac{(1 - p)^2}{p\mathcal{H} + (1 - p)} < 0,$$

$$\frac{\partial \mathcal{H}}{\partial \lambda} = \rho(\rho - 3 + 2\lambda) < 0,$$

and

$$\frac{\partial \mathcal{H}}{\partial \rho} = -\lambda[2(1 - \rho) + (1 - \lambda)] < 0.$$

Thus

$$\frac{dV_H^*}{d\lambda} = \frac{\partial V_H^*}{\partial \mathcal{H}} \frac{\partial \mathcal{H}}{\partial \lambda} > 0$$

and

$$\frac{dV_H^*}{d\rho} = \frac{\partial V_H^*}{\partial \mathcal{H}} \frac{\partial \mathcal{H}}{\partial \rho} > 0. \quad \square$$

B Additional Results

B.1 Continuous Distribution of Quality

Suppose the quality of the policy available to a high-ability policymaker in period t is drawn from a distribution with density $f(q_t)$, where $f(q_t) > 0$ for all $q_t \in [0, 1]$. Suppose further that if a policymaker implements a policy with quality q_t , it succeeds in each remaining period with probability $g(q_t)$, where $g(q_t)$ is strictly increasing in q_t , $g(0) = 0$, and $g(1) = 1$. The rest of the model is identical to the baseline model where $b = 0$.

To summarize, the sequence of play is as follows:

0. Nature draws the politician's type $\theta \in \{H, L\}$ and a policy-quality sequence $(q_1, q_2) \in [0, 1]^2$.
1. The politician observes (θ, q_1) and chooses $a_1 \in \{0, 1\}$. If $a_1 = 0$, the game proceeds to (3).
2. If $a_1 = 1$, the policy succeeds with probability $g(q_1)$; otherwise, $s_1 = 0$ and the game proceeds to (4).
3. The politician observes q_2 and chooses $a_2 \in \{0, 1\}$. If $a_2 = 0$, the game proceeds to (5).
4. If a policy has been implemented and has not yet succeeded, it succeeds with probability $g(q_t)$; otherwise, $s_2 = 0$.
5. The voter observes $(a_1, a_2, \hat{s})$, forms beliefs, and payoffs are realized.

Proposition 9. *There exists an equilibrium in which:*

1. *the high type implements the policy in period 1 if and only if $q_1 \geq z^*$,*
2. *the high type implements the policy in period 2 for all q_2 ,*

and the low type implements in period 1 with probability $\gamma_1^ \in (0, 1)$ and in period 2 with probability $1 - \gamma_1^*$.*

Proof: We construct an equilibrium of the proposed form and verify that strategies and beliefs satisfy sequential rationality and Bayes' rule.

Step 1: Distribution of no-success information sets.

Fix a cutoff $z \in (0, 1)$. Suppose the high type implements in period 1 if and only if $q_1 \geq z$, and implements in period 2 for all q_2 following delay. Then, the high type induces the following distribution over no-success information sets:

$$h_1(z) = \int_z^1 (1 - g(q))^2 dF(q_1),$$

$$h_2(z) = F(z) \int_0^1 (1 - g(q)) dF(q_2),$$

where $h_1(z)$ is the probability of early implementation with no success and $h_2(z)$ is the probability of late implementation with no success.

Step 2: Low-type behavior.

Let the low type implement in period 1 with probability γ_1 and in period 2 with probability $1 - \gamma_1$. Let

$$\mu_1(z) = \frac{ph_1(z)}{ph_1(z) + (1 - p)\gamma_1},$$

$$\mu_2(z) = \frac{ph_2(z)}{ph_2(z) + (1 - p)(1 - \gamma_1)}$$

denote posterior beliefs following unsuccessful early and late implementation.

Indifference of the low type requires $\mu_1(z) = \mu_2(z)$. This implies

$$\frac{h_1(z)}{\gamma_1} = \frac{h_2(z)}{1 - \gamma_1},$$

so that

$$\gamma_1(z) = \frac{h_1(z)}{h_1(z) + h_2(z)}.$$

Since $h_1(z), h_2(z) > 0$ for all $z \in (0, 1)$, it follows that $\gamma_1(z) \in (0, 1)$. Moreover, $\gamma_1(z)$ is decreasing in z , since $h_1(z)$ is decreasing and $h_2(z)$ is increasing in z .

Under this condition,

$$\mu_1(z) = \mu_2(z) \equiv \mu(z) = \frac{p}{p + \frac{1-p}{\mathcal{H}(z)}},$$

where $\mathcal{H}(z) = h_1(z) + h_2(z)$.

Step 3: High-type indifference.

Consider a high type with $q_1 = z$. If he implements in period 1, his expected payoff is

$$g(z)(2 - g(z)) + (1 - g(z)(2 - g(z)))\mu(z).$$

If he delays, his expected payoff is

$$\int_0^1 [g(q) + (1 - g(q))\mu(z)] dF(q_t).$$

Define

$$\omega(z) \equiv g(z)(2 - g(z)) + (1 - g(z)(2 - g(z)))\mu(z) - \int_0^1 [g(q) + (1 - g(q))\mu(z)] dF(q_t),$$

and

$$\bar{g} \equiv \int_0^1 g(q) dF(q_t).$$

Substituting $\mu(z)$ and $\bar{g}$ into $\omega(z)$, we have

$$\omega(z) = (g(z)(2 - g(z)) - \bar{g})(1 - \mu(z)).$$

Since $\mu(z) < 1$ for all $z \in (0, 1)$, the sign of $\omega(z)$ is determined by the sign of $g(z)(2 - g(z)) - \bar{g}$.

Define

$$\psi(z) \equiv g(z)(2 - g(z)).$$

Because g is continuous and strictly increasing with $g(0) = 0$ and $g(1) = 1$, we have

$$\psi(0) = 0, \quad \psi(1) = 1,$$

and

$$\psi'(z) = 2g'(z)(1 - g(z)) > 0$$

for all $z \in (0, 1)$. Hence ψ is continuous and strictly increasing on $[0, 1]$.

Moreover, $0 < \bar{g} < 1$, since $f(q_t) > 0$ on $[0, 1]$ and g is not constant. Therefore,

$$\psi(0) - \bar{g} < 0 \quad \text{and} \quad \psi(1) - \bar{g} > 0.$$

By continuity of ψ , there exists a unique $z^* \in (0, 1)$ such that

$$g(z^*)(2 - g(z^*)) = \bar{g}.$$

At this value, $\omega(z^*) = 0$, so the high type is indifferent between implementing in period 1 and delaying when $q_1 = z^*$.

Step 4: Off-path beliefs.

Consider a deviation to no implementation. The low type has a strictly stronger incentive to deviate than the high type, since he cannot generate success. Under D1, beliefs assign probability zero to the high type following such deviations. Hence, the posterior following no implementation is zero, and neither type benefits from deviating.

Thus, there exists an equilibrium characterized by a cutoff $z^* \in (0, 1)$ and mixing probability $\gamma_1^* = \gamma_1(z^*) \in (0, 1)$, completing the proof. $\square$

Proposition 10. *The odds the low type implements early are lower than the odds the high type implements early.*

Proof: Given the equilibrium strategies, the odds that the high type implements early, conditional on implementing in either period, are $\frac{1-F(z^*)}{F(z^*)}$, while the odds that the low type implements early are $\frac{\gamma_1}{1-\gamma_1}$.

Using the low type's indifference condition, it follows that

$$\frac{\gamma_1}{1-\gamma_1} = \frac{h_1(z^*)}{h_2(z^*)}.$$

Thus, the desired inequality holds if and only if

$$\frac{1-F(z^*)}{F(z^*)} > \frac{h_1(z^*)}{h_2(z^*)},$$

or equivalently,

$$(1-F(z^*)) \int_0^1 (1-g(q)) dF(q_t) > \int_{z^*}^1 (1-g(q))^2 dF(q_t). \quad (12)$$

Since g is strictly increasing, $1-g(q)$ is strictly decreasing on $[0, 1]$. Hence, the average of $1-g(q)$ over the entire support strictly exceeds its average over the upper tail $[z^*, 1]$, so

$$\int_0^1 (1-g(q)) dF(q_t) > \frac{1}{1-F(z^*)} \int_{z^*}^1 (1-g(q)) dF(q_t). \quad (13)$$

Moreover, for all $q \in [0, 1]$,

$$(1-g(q))^2 \leq 1-g(q),$$

with strict inequality on any set where $g(q) \in (0, 1)$. Since $f(q_t) > 0$ on $[0, 1]$ and $z^* < 1$, this set has positive measure in $[z^*, 1]$, implying

$$\int_{z^*}^1 (1 - g(q))^2 dF(q_t) < \int_{z^*}^1 (1 - g(q)) dF(q_t). \quad (14)$$

Combining (13) and (14),

$$(1 - F(z^*)) \int_0^1 (1 - g(q)) dF(q_t) > \int_{z^*}^1 (1 - g(q))^2 dF(q_t),$$

which is equivalent to (12). $\square$

Proposition 11. *Policies implemented early have higher expected quality than policies implemented late:*

$$\mathbb{E}[q \mid a_1 = 1] > \mathbb{E}[q \mid a_1 = 0, a_2 = 1].$$

Proof:

Given the high type's and low type's equilibrium strategies, expected policy quality in the first period is given by

$$\mathbb{E}[q_1 \mid \theta = 1 \& a_1 = 1] = \frac{p(1 - F(z^*))}{p(1 - F(z^*)) + (1 - p)\gamma_1},$$

while expected policy quality in the second period is given by

$$\mathbb{E}[q_2 \mid \theta = 1 \& a_1 = 0, a_2 = 1] = \frac{pF(z^*)}{pF(z^*) + (1 - p)\gamma_2}$$

Since the high type implements early if and only if $q_1 \geq z^*$ and implements late for all q_2 , it follows that

$$\mathbb{E}[q_1 \mid \theta = 1 \& a_1 = 1] = \frac{1}{1 - F(z^*)} \int_{z^*}^1 q_1 dF(q_1) > \int_0^1 q_2 dF(q_2) = \mathbb{E}[q_2 \mid \theta = 1 \& a_1 = 0, a_2 = 1].$$

Moreover, from the previous result,

$$\frac{1 - F(z^*)}{F(z^*)} > \frac{\gamma_1}{1 - \gamma_1}.$$

This implies

$$\frac{p(1 - F(z^*))}{p(1 - F(z^*)) + (1 - p)\gamma_1} > \frac{pF(z^*)}{pF(z^*) + (1 - p)(1 - \gamma_1)},$$

which establishes the result. $\square$